

Strict hot spots and absence of interior critical points on smooth simply connected planar domains

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Abstract

This manuscript presents a readability revision of a proposed proof of the strict hot spots conjecture for smooth bounded simply connected planar domains. The claimed result is that every nonzero eigenfunction for the first positive Neumann eigenvalue has nonvanishing gradient in the interior, so all its global maxima and minima lie on the boundary, including when the eigenvalue is multiple. George Coote worked with GPT 6.1 Sol to expand and clarify the supplied argument so that he and other readers can review it with less effort. This revision does not constitute independent verification of the claimed result.

1 Introduction

This document revises the exposition of a proposed proof in the supplied manuscript. George Coote's purpose in working with GPT 6.1 Sol was to make that argument readable enough for him and others to examine its mathematical validity without having to reconstruct numerous omitted steps. The main statements and proof strategy were already present in the supplied text. The revision expands definitions, calculations, references, and logical transitions; it does not claim that Coote originated the result or its proof, or that the proposed proof has been independently verified. The following introduction describes the claim and the argument presented for it.

The hot-spots problem arose from Rauch's discussion of the long-time behavior of heat flow with insulating boundary conditions [33]. Its spectral formulation asks whether the extrema of a first-positive Neumann eigenfunction occur on the boundary. Bañuelos and Burdzy distinguished strict and nonstrict formulations, and assertions about every eigenfunction from the existence of one favorable eigenfunction [7]. These distinctions matter when the eigenvalue is multiple.

Let $\Omega \subset \mathbb{R}^2$ be a nonempty bounded simply connected open set with C^∞ boundary. In particular, Ω is connected. Set

$$\lambda_N = \inf_{\substack{0 \neq v \in H^1(\Omega; \mathbb{R}) \\ \int_{\Omega} v = 0}} \frac{\int_{\Omega} |\nabla v|^2}{\int_{\Omega} v^2}, \quad (1.1)$$

and let V be the real Neumann eigenspace for λ_N . This is the first positive eigenvalue; sources that count the constant eigenfunction first call it the second Neumann eigenvalue.

The supplied manuscript claims the following theorem.

Theorem 1.1. *For every $0 \neq u \in V$,*

$$\nabla u(x) \neq 0 \quad (x \in \Omega).$$

In particular,

$$\min_{y \in \partial\Omega} u(y) < u(x) < \max_{y \in \partial\Omega} u(y) \quad (x \in \Omega). \quad (1.2)$$

The theorem imposes no convexity or symmetry condition. It applies to every member of the first positive eigenspace, including when the eigenvalue is multiple. The nonvanishing-gradient conclusion also rules out interior saddle points.

Background and methods. Topology matters in the hot-spots problem. Burdzy and Werner constructed a planar counterexample with two holes and a simple first-positive eigenvalue [16]. Bass and Burdzy obtained examples with both extrema strictly in the interior [14], and Burdzy subsequently obtained this behavior with just one hole [15]. The simply connected planar conjecture is stated explicitly in [15, Conjecture 1.2(ii)].

Several geometric special cases already establish strong gradient conclusions. Jerison and Nadirashvili proved a positive directional derivative for every first-positive eigenfunction on convex planar domains with two orthogonal reflection symmetries, allowing multiple eigenvalues [23, Theorem 1.4]. Probabilistic coupling methods developed by Bañuelos and Burdzy [7] led to further results. Pascu proved boundary-only extrema for antisymmetric eigenfunctions on $C^{1,\alpha}$ convex planar domains with one reflection symmetry, where $0 < \alpha < 1$ [29]. Atar and Burdzy treated every first-positive eigenfunction on lip domains, a class of Lipschitz domains bounded between graphs of functions with Lipschitz constants at most one [13]. Miyamoto obtained interior critical-point exclusion under a spectral-diameter condition, giving nonsymmetric nearly circular convex examples [28, Lemma 1.2 and Theorem A].

For polygons, Siudeja proved hot-spots results for a class of acute triangles [36]. Judge and Mondal, with their subsequent erratum, excluded interior critical points for every first-positive Neumann eigenfunction on every Euclidean triangle [24, 25]. Chen, Gui and Yao established the finer classification of nonvertex critical points [17]. These results concern domains with corners; Theorem 1.1 concerns smooth simply connected planar domains.

Recent quantitative work measures how far the hot-spots conclusion can fail. For general bounded connected Lipschitz domains, de Dios Pont, Hsu and Taylor determined the sharp dimension-dependent upper bound for the ratio of a first-positive Neumann eigenfunction's supremum over the domain to its maximum on the boundary [18, Definition 1 and Theorem 4]. Deng, Jiang and Yang obtained quantitative bounds for convex domains in two-dimensional space forms [19]. Such ratios concern extrema; excluding every interior critical point also requires excluding saddle points.

The supplied proof starts from Rohleder's variational principle for tangent vector fields [9, 34]. In its lip-domain application, taking componentwise absolute values after a suitable rotation preserves the required tangent boundary condition and yields directional monotonicity. Its argument uses scalar boundary multipliers to keep tangency on an arbitrary smooth boundary, constructing those multipliers through a reciprocal Green-kernel theorem.

Each Cartesian derivative of an eigenfunction solves the same Helmholtz equation, but its boundary values need not have a fixed sign. The Neumann condition constrains the two derivatives together: their vector is tangent to the boundary. The argument uses that constraint through a nonnegative quadratic form on tangent boundary data. Its nullspace consists exactly of the

boundary traces of first-positive eigenfunction gradients. The kernel construction supplies scalar boundary multipliers that preserve this nullspace if an interior critical point exists.

The mechanism. Write $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and $S^1 = \partial\mathbb{D}$, and let $\Phi : \overline{\mathbb{D}} \rightarrow \overline{\Omega}$ be a smooth conformal parametrization. For a tangent vector field X , the divergence–curl energy $Q(X)$ is at least $\lambda_N \|X\|_2^2$, with equality exactly for gradients of functions in V ; see Proposition 3.1. After the conformal change of variables, solving the scalar Helmholtz equation in each fixed Cartesian component extends Lipschitz boundary data $h : S^1 \rightarrow \mathbb{R}^2$ to a vector field X_h on Ω . For tangent data, the quadratic form

$$E(h) = Q(X_h) - \lambda_N \|X_h\|_2^2$$

is nonnegative, and it vanishes exactly when $X_h = \nabla v$ for some $v \in V$. Here tangency at s means tangency to $\partial\Omega$ at $\Phi(s)$. For a nonzero eigenfunction $u \in V$, put $g(s) = \nabla u(\Phi(s))$. This is the gradient in the coordinates of Ω , parametrized on the circle. Its extension is $X_g = \nabla u$, so $E(g) = 0$, and a real Lipschitz scalar multiplier b satisfies

$$E(bg) = \frac{1}{2} \iint_{S^1 \times S^1} N(s, t) (b(s) - b(t))^2 g(s) \cdot g(t) \, ds \, dt.$$

Here N is a symmetric boundary kernel, strictly positive off the diagonal, and the corresponding interior evaluation kernel is denoted by K ; see Proposition 4.4.

The sign of $g(s) \cdot g(t)$ prevents a direct positivity argument in this identity. The kernel theorem supplies multipliers whose squared differences remove the factor N . For every interior point p ,

$$D_p(s, t) = \frac{K(p, s)K(p, t)}{N(s, t)} \quad (s \neq t), \quad D_p(s, s) = 0,$$

is conditionally negative semidefinite. Lemma 6.1 constructs a separable real Hilbert space and an injective Lipschitz map b from the circle into that space, with squared distances D_p . The construction uses the positive anchored kernel, a quotient by the zero-norm subspace, and completion; the kernel estimates give injectivity and the Lipschitz bound. Applying the energy identity to the coordinates b_j of b in an orthonormal basis, and justifying the sum by dominated convergence, gives

$$\sum_j E(b_j g) = \frac{1}{2} |\nabla u(\Phi(p))|^2.$$

At a critical point every nonnegative summand vanishes. The equality case then gives $v_j \in V$ with boundary gradient $b_j g$. If the set $S_g = \{s : g(s) \neq 0\}$ is dense in the circle, injectivity of b produces three linearly independent eigenfunctions, contradicting Proposition 2.5. If S_g is not dense, the gradients vanish on an open boundary arc. Two independent eigenfunctions then have a nonzero linear combination with both zero value and zero normal derivative on that arc. Its local extension by zero solves the Helmholtz equation across the arc, and interior uniqueness gives the contradiction.

The construction of D_p is the central analytic step. It first regularizes the disk Green function so that the entrywise reciprocal of every finite kernel matrix has at most one positive eigenvalue. This property survives successive rank-one updates along columns of the current kernel matrix, which sum the Green resolvent over a finite set of integration nodes. Positive quadrature and monotone limits then pass from those matrices to the singular Green kernel and finally to its boundary kernels. Taking the boundary limit before removing a compact-support cutoff avoids treating a Poisson kernel as an L^2 function when it need not be one.

The kernel theorem, Theorem 5.1, holds for every bounded nonnegative disk potential satisfying a weighted Dirichlet inequality with constant $d < 1$, independently of the conformal origin of that potential. The conformal potential satisfies this condition by equation (4.2). The reciprocal-matrix condition is related to the McCullough–Quiggin characterization of complete Nevanlinna–Pick kernels [26, 27, 31, 32]; see also [10, Corollary 1.12]. The condition used here requires at most one positive eigenvalue in each finite sampled reciprocal matrix, allowing zero diagonal entries in the limiting matrix. This condition concerns finite matrices, rather than reciprocals of integral operators. Schoenberg’s squared-distance correspondence underlies the Hilbert-space construction proved in lemma 6.1 [35]. The analytic work here concerns the singular Green kernel, its positive-potential resolvent, and its boundary limit.

Organization and background. Section 2 proves the spectral gap, the smooth conformal reduction, and the multiplicity bound. Section 3 proves the vector variational principle. Section 4 constructs the boundary kernels and derives the multiplier identity. Section 5 proves their conditional negativity, and Section 6 completes the argument. The argument relies on Sobolev compactness, Poincaré and trace inequalities on smooth bounded domains, Poisson solvability and elliptic boundary regularity, the maximum and boundary point principles, elementary complex analysis, and planar separation. The later sections give detailed arguments for the specialized ingredients. The smooth-boundary regularity input is [8, Chapter III, Theorems 6.1, 6.4 and 6.5]: L^2 Poisson data give H^2 solutions under homogeneous Dirichlet or Neumann conditions, and higher regularity follows with smoother data. Subtracting a smooth extension handles the inhomogeneous Dirichlet data used below. Bootstrap and Sobolev embedding make weak Neumann eigenfunctions smooth up to the boundary. Interior solutions of $(\Delta + \lambda_N)v = 0$ are analytic; alternatively, $e^{\sqrt{\lambda_N}t}v(x)$ is harmonic in one additional variable, as shown in Fact 2.7.

Human revision and use of GPT 6.1 Sol. The exposition in sections 2 to 6 was developed through a passage-by-passage revision directed by George Coote, using OpenAI’s GPT 6.1 Sol. The aim was to reduce the effort needed to understand and review the supplied proof. Coote selected the scope and pace of the work, identified compressed or ambiguous arguments, challenged terminology and unsupported transitions, and decided which explanations and calculations should enter the text. The revision consistently uses λ_N for the first positive Neumann eigenvalue and makes the test spaces, trace conditions, equality cases, and dependencies between results explicit.

In section 2, this scrutiny led to fuller explanations of variational equality and plane-wave independence, smooth conformal boundary regularity, the boundary-sign argument, and the multiplicity bound. Coote required the planar separation argument to rest on a precise crosscut statement and proof, with the figure serving as an illustration. In section 3, the revisions explain the construction and compatibility of the Neumann and Dirichlet potentials, the truncation and weak-gradient arguments, Sobolev integration by parts, and the equality case. They also expand the curvature identity and its passage from smooth fields to Sobolev data.

In section 4, Coote requested explicit conformal changes of variables, weak solution and uniqueness arguments, kernel bounds, and justifications for exchanging integrals and sums. The revised energy calculation identifies the componentwise extensions, the Fourier normalization, curvature cancellation, and absolute integrability in the multiplier identity. In section 5, Coote required a clear distinction between finite sampled matrices and integral operators, a literature comparison for the reciprocal condition, and explicit quadrature nodes, weights, matrix updates, and convergence arguments. The revision separates diagonal from off-diagonal boundary limits and

explains the compact-support cutoff before its removal. In section 6, Coote requested the Hilbert-space construction, pointwise boundary gradient identities, the transfer of linear independence to eigenfunctions, and the integration-by-parts calculation for local zero extension. The final extrema argument explicitly uses the extreme value theorem and Fermat's stationary-point theorem from elementary calculus: at an interior local maximum or minimum, a differentiable function has zero gradient.

GPT 6.1 Sol supplied much of the new mathematical prose, detailed derivations, supporting calculations, reference work, and LaTeX implementation in response to these interventions. Coote reviewed the proposed explanations, controlled their level of detail, and authorized successive additions. This account summarizes the section-by-section contribution records accompanying the source files; it describes the development of the revised exposition. The author line identifies Coote's work on this revision with GPT 6.1 Sol; it is not a claim that he discovered or proved the underlying result. The recorded mathematical checks were model-assisted reviews of the arguments. The purpose of making those arguments explicit is to enable further mathematical review, and their correctness remains to be assessed.

2 Spectral and planar preliminaries

Let $\Omega \subset \mathbb{R}^2$ be a nonempty bounded connected open set with C^∞ boundary. Simple connectivity is not needed for the spectral comparison below; it will be assumed explicitly in lemma 2.2. All integrals over Ω are with respect to Lebesgue measure. Write

$$\lambda_D = \inf_{0 \neq v \in H_0^1(\Omega; \mathbb{R})} \frac{\int_\Omega |\nabla v|^2}{\int_\Omega v^2}, \quad \lambda_N = \inf_{\substack{0 \neq v \in H^1(\Omega; \mathbb{R}) \\ \int_\Omega v = 0}} \frac{\int_\Omega |\nabla v|^2}{\int_\Omega v^2}.$$

Here λ_N denotes the first *positive* Neumann eigenvalue, denoted by μ in the source manuscript; the constant eigenfunctions have eigenvalue zero. The Poincaré inequalities give $\lambda_D, \lambda_N > 0$. Compactness of the Sobolev embeddings into $L^2(\Omega)$ gives attainment of these infima: normalize a minimizing sequence in L^2 , take a weakly convergent subsequence in the relevant Sobolev space and a strongly convergent subsequence in L^2 , and use weak lower semicontinuity of the gradient norm.

We record the facts used in the proof. The spaces of complex-valued functions are understood componentwise: $H^1(\Omega; \mathbb{C}) = H^1(\Omega; \mathbb{R}) + iH^1(\Omega; \mathbb{R})$, and similarly for H_0^1 .

Fact 2.1 (Absolute values and traces). For real $v \in H^1(\Omega)$, one has $|v| \in H^1(\Omega)$ and

$$\nabla |v| = \operatorname{sgn}(v) \nabla v \quad \text{almost everywhere,} \quad \operatorname{sgn}(0) = 0.$$

Also $\nabla v = 0$ almost everywhere on $\{v = 0\}$, so $|\nabla |v|| = |\nabla v|$ almost everywhere. The trace operator $\operatorname{Tr} : H^1(\Omega) \rightarrow L^2(\partial\Omega)$ is the bounded extension of boundary restriction for smooth functions, and

$$H_0^1(\Omega) = \{v \in H^1(\Omega) : \operatorname{Tr} v = 0\}.$$

Traces commute with absolute values, so $v \in H_0^1(\Omega)$ implies $|v| \in H_0^1(\Omega)$. In particular, replacing a real Dirichlet minimizer by its absolute value preserves its Rayleigh quotient.

For the weak derivative formula see [1, Proposition 3.22, equation (3.7), pp. 57–58]; the positive- and negative-part formulas immediately following it imply that $\nabla v = 0$ almost everywhere on

$\{v = 0\}$. For traces and the characterization of zero trace, see [1, Theorem 3.44, p. 72, and Section 3.11, p. 75], using $p = 2$ and boundary charts to pass from the half-space to Ω . Compatibility with absolute values follows by smooth approximation, continuity of the trace, and the absolute-value operation in H^1 .

Fact 2.2 (Neumann variational equality over $\mathbb{C}$). Define

$$a_N(f, g) = \int_{\Omega} \nabla f \cdot \overline{\nabla g} - \lambda_N \int_{\Omega} f \bar{g}.$$

For $w \in H^1(\Omega; \mathbb{C})$ with $\int_{\Omega} w = 0$, one has $a_N(w, w) \geq 0$. Equality holds if and only if w satisfies

$$\int_{\Omega} \nabla w \cdot \overline{\nabla \psi} = \lambda_N \int_{\Omega} w \bar{\psi} \quad (\psi \in H^1(\Omega; \mathbb{C})). \quad (2.1)$$

Proof. Write $w = a + ib$ with a, b real and mean zero. Then

$$a_N(w, w) = \left(\int_{\Omega} |\nabla a|^2 - \lambda_N \int_{\Omega} a^2 \right) + \left(\int_{\Omega} |\nabla b|^2 - \lambda_N \int_{\Omega} b^2 \right) \geq 0.$$

Thus the complex and real Rayleigh infima coincide. If $a_N(w, w) = 0$, for every mean-zero $\eta \in H^1(\Omega; \mathbb{C})$ and $t \in \mathbb{R}$,

$$0 \leq a_N(w + t\eta, w + t\eta) = 2t \operatorname{Re} a_N(w, \eta) + t^2 a_N(\eta, \eta).$$

Both signs of arbitrarily small t give $\operatorname{Re} a_N(w, \eta) = 0$; replacing η by $i\eta$ gives $\operatorname{Im} a_N(w, \eta) = 0$ as well. For an arbitrary ψ , subtract its mean to obtain such an η . The remaining constant c satisfies $a_N(w, c) = -\lambda_N \bar{c} \int_{\Omega} w = 0$. This proves equation (2.1). Conversely, testing that equation with $\psi = w$ gives equality. This is the weak Neumann eigenfunction equation: the variational formulation of the Laplacian eigenvalue problem with homogeneous Neumann boundary conditions. The test space is all of $H^1(\Omega)$, rather than $H_0^1(\Omega)$. Elliptic regularity on our smooth domain, using [8, Chapter III, Theorems 6.4 and 6.5], gives $-\Delta w = \lambda_N w$ in Ω and $\partial_{\nu} w = 0$ on $\partial\Omega$. In particular, the complex solution space is the complexification of the real eigenspace: separating real and imaginary parts in the weak equation shows that both parts are real eigenfunctions (or zero). $\square$

Fact 2.3 (Finite-dimensional Neumann eigenspaces). For any fixed $\lambda \geq 0$, the complex vector space of weak Neumann eigenfunctions with eigenvalue λ is finite-dimensional.

Proof. Otherwise choose an L^2 -orthonormal sequence u_j in that eigenspace. Testing each weak eigenfunction equation by u_j gives $\|u_j\|_{H^1}^2 = 1 + \lambda$. Compactness of $H^1(\Omega) \hookrightarrow L^2(\Omega)$ then gives an L^2 -convergent subsequence, whereas orthonormality gives $\|u_j - u_{\ell}\|_{L^2} = \sqrt{2}$ for $j \neq \ell$, a contradiction. $\square$

See also [2, Theorem 8, p. 6], for finite multiplicity of Dirichlet and Neumann eigenvalues under the compact-resolvent assumptions satisfied by bounded smooth domains. If the real eigenspace has real dimension d , its complexification has complex dimension d .

Fact 2.4 (Independence of plane waves). For distinct $k_1, \dots, k_m \in \mathbb{R}^2$, the functions $x \mapsto e^{ik_j \cdot x}$ are linearly independent over $\mathbb{C}$ on every nonempty open subset of $\mathbb{R}^2$.

Proof. Suppose $\sum_j a_j e^{ik_j \cdot x} = 0$ on such a set, and choose an interior ball $B(x_0, r)$. An almost-everywhere identity is also a pointwise identity by continuity. Choose a unit vector d with $(k_j - k_\ell) \cdot d \neq 0$ for every $j \neq \ell$: each pair excludes only two directions on the unit circle. On $x = x_0 + td$, $|t| < r$, the identity is

$$\sum_{j=1}^m b_j e^{i\omega_j t} = 0, \quad b_j = a_j e^{ik_j \cdot x_0}, \quad \omega_j = k_j \cdot d.$$

The ω_j are distinct. Differentiating $q = 0, \dots, m-1$ times at $t = 0$ gives $\sum_j b_j (i\omega_j)^q = 0$. The determinant of this Vandermonde system is $\prod_{j < \ell} (i\omega_\ell - i\omega_j) \neq 0$. Thus every b_j , and hence every a_j , is zero. $\square$

For the general independence of distinct characters, including the exponentials $t \mapsto e^{i\omega t}$ on the additive group $\mathbb{R}$, see [3, Theorem 2.1, p. 1]. The proof above establishes the local version on an interval that is needed here.

The strict comparison below is the first-mode case of the Friedlander–Filonov comparison [21, 20]. The proof uses Filonov’s plane-wave argument, with the equality and strictness steps expanded.

Lemma 2.1. *One has $\lambda_N < \lambda_D$.*

Proof. Choose a real nonzero Dirichlet minimizer and replace it by its absolute value using Fact 2.1. The resulting $\phi \in H_0^1(\Omega; \mathbb{R})$ satisfies

$$\phi \geq 0, \quad \int_{\Omega} |\nabla \phi|^2 = \lambda_D \int_{\Omega} \phi^2, \quad \int_{\Omega} \phi > 0.$$

The last inequality uses only nonnegativity and nonzero L^2 norm; pointwise strict positivity is not needed.

We now allow complex-valued trial functions, keeping the same real variational numbers λ_N and λ_D . For $k \in \mathbb{R}^2$ with $|k|^2 = \lambda_D$, put

$$e_k(x) = e^{ik \cdot x}, \quad c_k = \frac{\int_{\Omega} e_k}{\int_{\Omega} \phi}, \quad w_k = e_k - c_k \phi.$$

Then $w_k \in H^1(\Omega; \mathbb{C})$ and $\int_{\Omega} w_k = 0$. Define

$$B_D(f, g) = \int_{\Omega} \nabla f \cdot \overline{\nabla g} - \lambda_D \int_{\Omega} f \overline{g}, \quad Q_D(f) = B_D(f, f).$$

Since $\nabla e_k = i k e_k$ and $|e_k| = 1$, $Q_D(e_k) = 0$; the minimizing identity gives $Q_D(\phi) = 0$. Furthermore,

$$B_D(e_k, \phi) = 0.$$

Indeed $-\Delta e_k = \lambda_D e_k$, and testing this equation against $\phi \in H_0^1(\Omega)$ gives the displayed identity. Rigorously, first test against compactly supported smooth functions and pass to ϕ by density in H_0^1 ; equivalently, the integration-by-parts boundary term vanishes because $\text{Tr } \phi = 0$. Expanding the quadratic form therefore gives

$$Q_D(w_k) = Q_D(e_k) + |c_k|^2 Q_D(\phi) - 2 \text{Re}(\overline{c_k} B_D(e_k, \phi)) = 0.$$

Every nonzero w_k consequently has Rayleigh quotient λ_D .

To see that there are nonzero w_k , and to prepare the strictness argument, choose any m distinct vectors $k_1, \dots, k_m$ on the circle $\{|k|^2 = \lambda_D\}$. Independence of the plane waves (Fact 2.4) and $e_{k_j} = w_{k_j} + c_{k_j}\phi$ give

$$m = \dim_{\mathbb{C}} \text{span}\{e_{k_1}, \dots, e_{k_m}\} \leq \dim_{\mathbb{C}} \text{span}\{w_{k_1}, \dots, w_{k_m}\} + 1.$$

In particular $m = 2$ supplies a nonzero mean-zero trial function, so the complex Neumann inequality in Fact 2.2 implies $\lambda_N \leq \lambda_D$.

Suppose equality held. Each nonzero w_k would attain equality in the Neumann inequality and hence, by Fact 2.2, satisfy equation (2.1); the zero functions satisfy that equation as well. Thus all w_k lie in the fixed complex Neumann eigenspace for λ_N . But the preceding dimension estimate gives a span of dimension at least $m - 1$ for every m . There are arbitrarily many distinct k on the circle because $\lambda_D > 0$, so this eigenspace would be infinite-dimensional, contrary to Fact 2.3. Therefore $\lambda_N < \lambda_D$. $\square$

Identify $\mathbb{R}^2$ with $\mathbb{C}$, and write $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and $S^1 = \partial\mathbb{D}$. Smoothness up to the boundary means smoothness in local coordinates on the manifold with boundary $\bar{\Omega}$; in particular, all derivatives extend continuously to the boundary. A boundary chart is a smooth local diffeomorphism $F(s, t)$ that takes $t = 0$ to $\partial\Omega$ and $t > 0$ to Ω , thereby flattening the boundary.

Fact 2.5 (Hopf boundary point lemma). Let $U \subset \mathbb{R}^2$ be open and let $u \in C^2(U) \cap C^1(\bar{U})$ satisfy $\Delta u \geq 0$ and $u < M$ in U . Suppose $p \in \partial U$, $u(p) = M$, and U satisfies the interior sphere condition at p : there is a ball $B_r(b) \subset U$ with $p \in \partial B_r(b)$. Then

$$\nabla u(p) \cdot \frac{p - b}{r} > 0.$$

If ∂U is smooth near p , the displayed vector is its outward unit normal, so this conclusion is $\partial_\nu u(p) > 0$.

See [1, Lemma 2.21, p. 29]; Definition 2.19 and the discussion preceding the lemma note that C^2 boundaries satisfy the interior sphere condition. The result can be applied locally on an interior tangent ball, so the function need only be defined near the boundary point in question.

For the classical boundary-regularity theory of conformal maps, see [30]. The following proof constructs the smooth parametrization needed here.

Lemma 2.2. *If Ω is simply connected, there is a conformal bijection $\Phi : \mathbb{D} \rightarrow \Omega$ extending to a smooth diffeomorphism $\bar{\mathbb{D}} \rightarrow \bar{\Omega}$. Its complex derivative Φ' extends smoothly and is nonzero everywhere on $\bar{\mathbb{D}}$.*

Proof. Fix $a \in \Omega$. First construct the harmonic function l with boundary values $\log|z - a|$. Since a has positive distance from $\partial\Omega$, these boundary data are smooth. More explicitly, choose a smooth cutoff χ equal to one near $\partial\Omega$ and zero near a . The function $G(z) = \chi(z) \log|z - a|$, defined to be zero near a , belongs to $C^\infty(\bar{\Omega})$. Since $\Delta G \in L^2(\Omega) \subset H^{-1}(\Omega)$, the Dirichlet existence theorem [1, Theorem 4.11, p. 99] gives a unique $v \in H_0^1(\Omega)$ satisfying

$$\int_{\Omega} \nabla v \cdot \nabla \eta = \int_{\Omega} (\Delta G) \eta \quad (\eta \in H_0^1(\Omega)).$$

Here $H^{-1}(\Omega)$ is the continuous dual of $H_0^1(\Omega)$; the inclusion of L^2 follows from Cauchy–Schwarz and the Poincaré inequality. Existence and uniqueness follow from the Riesz representation theorem (or Lax–Milgram) for the coercive Dirichlet energy form. This weak equation is $-\Delta v = \Delta G$. Higher-order boundary regularity [1, Theorem 4.31, p. 116] gives $v \in H^{k+2}(\Omega)$ for every integer $k \geq 0$, and Sobolev embedding [1, Corollary 4.32, p. 116] gives $v \in C^\infty(\overline{\Omega})$. Thus $l = G + v$ is harmonic and smooth up to the boundary, with the required boundary values.

The one-form $-l_y dx + l_x dy$ is closed because $\Delta l = 0$. Simple connectivity makes it exact, so it has a real potential $\tilde{l}$ with

$$\tilde{l}_x = -l_y, \quad \tilde{l}_y = l_x.$$

Consequently $l + i\tilde{l}$ is holomorphic on Ω . To check smoothness of the conjugate at the boundary, let $h(s, t) = \tilde{l}(F(s, t))$ in a boundary chart. The displayed identities and the chain rule show that $\partial_t h$ extends smoothly to $t = 0$. After shrinking the chart, fix an interior height $t_0 > 0$. The identity

$$h(s, t) = h(s, t_0) + \int_{t_0}^t \partial_t h(s, \tau) d\tau$$

gives a smooth extension to $t = 0$. These local extensions agree on overlaps by continuity and agreement in the interior. Hence $\tilde{l} \in C^\infty(\overline{\Omega})$.

Define

$$m(z) = (z - a) \exp(-l(z) - i\tilde{l}(z)).$$

This function is holomorphic in Ω , smooth up to the boundary, and has exactly one zero, namely a , of order one. The boundary values of l give $|m| = 1$ on $\partial\Omega$. The maximum modulus principle gives $|m| < 1$ throughout Ω : equality at an interior point would make m constant, contrary to its simple zero.

Near the boundary m is nonzero, and $H = \log |m| = \log |z - a| - l$ is harmonic, negative inside and zero on the boundary. The Hopf boundary point lemma (Fact 2.5), applied locally at each boundary point on an interior tangent ball small enough to avoid a , gives $\partial_\nu H > 0$, where ν is the outward unit normal. In particular $\nabla H \neq 0$ there. The Cauchy–Riemann equations imply $|\nabla H| = |m'|/|m|$ in the interior of this collar and, by continuity, on its boundary. Thus $m' \neq 0$ on $\partial\Omega$.

We next prove that m is bijective in the interior without appealing to the Riemann mapping theorem. The smooth collar neighbourhood theorem [4, Theorem 9.25, pp. 222–223] gives a smooth embedding

$$E : \partial\Omega \times [0, \varepsilon_0] \rightarrow \overline{\Omega}, \quad E(p, 0) = p,$$

with image a neighbourhood of the boundary and $E(p, t) \in \Omega$ for $t > 0$. Fix $w \in \mathbb{D}$. Continuity of m , compactness of $\partial\Omega$, and $|m| = 1$ there allow us to choose $0 < \varepsilon < \varepsilon_0$ such that

$$|m| > |w| \quad \text{on } E(\partial\Omega \times [0, \varepsilon]).$$

Shrink ε further so that this collar excludes a , and set

$$\Omega_\varepsilon = \Omega \setminus E(\partial\Omega \times (0, \varepsilon]).$$

This is a bounded open set with smooth boundary $E(\partial\Omega \times \{\varepsilon\})$, its closure is contained in Ω , and it contains a . Rouché’s theorem gives the same number of zeros of m and $m - w$ in Ω_ε , counted with multiplicity, because $|w| < |m|$ on its boundary. Here one may use the argument-principle version of Rouché’s theorem, summing over all oriented boundary components; no assumption

about the topology of Ω_ε is needed. The number is one, since m has only its simple zero at a . There are no zeros of $m - w$ in the omitted collar. Hence every $w \in \mathbb{D}$ has exactly one preimage in Ω , counted with multiplicity. This proves interior bijectivity and also $m' \neq 0$ in Ω : a zero derivative at z would give a zero of order at least two for $m - m(z)$. Thus the interior inverse is holomorphic.

To prove boundary surjectivity, fix $\zeta \in S^1$ and take $w_n \in \mathbb{D}$ converging to ζ . A subsequence of $z_n = m^{-1}(w_n)$ converges in the compact set $\overline{\Omega}$ to some z . Continuity gives $m(z) = \zeta$. Since $m(\Omega) \subset \mathbb{D}$, this z lies on $\partial\Omega$. Therefore $m(\partial\Omega) = S^1$.

We justify the local inverse at a boundary point p before proving boundary injectivity. Smooth maps in boundary charts admit local smooth extensions across $t = 0$. For such an extension of m , the Cauchy–Riemann equations at p , obtained by continuity from the interior, give

$$\det Dm(p) = |m'(p)|^2 > 0.$$

The real inverse function theorem therefore gives a local smooth diffeomorphism on a full neighbourhood of p . Its inverse maps nearby points of $\mathbb{D}$ into Ω , as follows. In a boundary chart, using the local extension, put

$$R(s, t) = |m(F(s, t))|^2 - 1.$$

Then $R(s, 0) = 0$ and $\partial_t R(s, 0) < 0$: the inward transverse vector $F_t(s, 0)$ has negative outward normal component, and the outward normal derivative of $|m|^2$ is positive by the Hopf conclusion. After shrinking the chart,

$$R(s, t) = tA(s, t), \quad A(s, t) = \int_0^1 \partial_t R(s, \theta t) d\theta < 0.$$

Thus $|m(F(s, t))| < 1$ holds precisely on the side $t > 0$ near p . This proves the stated property of the local inverse.

If distinct $p, q \in \partial\Omega$ had $m(p) = m(q) = \zeta$, choose disjoint local inverse neighbourhoods around them. Their images are open neighbourhoods of ζ in $\mathbb{C}$. The intersection contains a point of $\mathbb{D}$ sufficiently close to ζ , and the two local inverses produce distinct preimages in Ω . This contradicts interior injectivity. Hence m is also injective on the boundary. Interior and boundary images are disjoint, so m is a continuous bijection $\overline{\Omega} \rightarrow \overline{\mathbb{D}}$ and, by compactness, a homeomorphism. The local smooth inverses just constructed show that its inverse Φ is smooth up to the boundary. It is holomorphic in $\mathbb{D}$, and $\Phi' = 1/(m' \circ \Phi)$ there; this identity extends to the boundary by continuity. Since m' is nonzero everywhere on $\overline{\Omega}$, the asserted nonvanishing of Φ' follows. $\square$

For the remainder of this section assume that Ω is simply connected, and fix the map Φ from lemma 2.2. Write ds for arclength on S^1 , of total mass 2π , and $d\sigma$ for arclength on $\partial\Omega$. Let V be the real Neumann eigenspace for λ_N ; its members satisfy the variational formulation

$$\int_{\Omega} \nabla v \cdot \nabla \psi = \lambda_N \int_{\Omega} v \psi \quad (\psi \in H^1(\Omega; \mathbb{R})). \quad (2.2)$$

The full H^1 test space encodes the homogeneous Neumann boundary condition; compactly supported tests alone would encode only the interior Laplacian equation. They are smooth up to the boundary and satisfy $-\Delta v = \lambda_N v$ in Ω and $\partial_\nu v = 0$ on $\partial\Omega$. Taking $\psi = 1$ gives $\int_{\Omega} v = 0$, since $\lambda_N > 0$.

Lemma 2.3. *There is a fixed $\beta \in C^\infty(S^1)$, strictly positive everywhere, such that every $v \in V$ has boundary trace $h_v = (v \circ \Phi)|_{S^1}$ satisfying*

$$\int_{S^1} \beta h_v ds = 0.$$

If $v \neq 0$, then h_v is not identically zero and takes both positive and negative values.

Proof. Consider the symmetric bilinear form

$$b(f, g) = \int_{\Omega} \nabla f \cdot \nabla g - \lambda_N \int_{\Omega} fg \quad (f, g \in H_0^1(\Omega; \mathbb{R})).$$

The definition of λ_D gives the Dirichlet inequality

$$\int_{\Omega} |\nabla f|^2 \geq \lambda_D \int_{\Omega} f^2 \quad (f \in H_0^1(\Omega; \mathbb{R})). \quad (2.3)$$

By lemma 2.1, $\lambda_N < \lambda_D$, so

$$b(f, f) \geq \left(1 - \frac{\lambda_N}{\lambda_D}\right) \int_{\Omega} |\nabla f|^2 \geq c \|f\|_{H^1(\Omega)}^2$$

for some $c > 0$, using the Poincaré inequality in the last step. Thus b is coercive. It is also continuous, and $\eta \mapsto \int_{\Omega} \eta$ is a continuous linear functional on $H_0^1(\Omega)$. The Riesz representation theorem applied to the inner product b , or equivalently Lax–Milgram, gives a unique real $L \in H_0^1(\Omega)$ such that

$$b(L, \eta) = \int_{\Omega} \eta \quad (\eta \in H_0^1(\Omega; \mathbb{R})). \quad (2.4)$$

This is the weak Dirichlet problem

$$(-\Delta - \lambda_N)L = 1 \quad \text{in } \Omega, \quad \text{Tr } L = 0.$$

For this coercive solvability argument with a negative zeroth-order coefficient, see [1, Example 4.14, p. 100]; the displayed estimate verifies its hypothesis using the optimal Dirichlet constant. To justify smoothness, rewrite the equation as $-\Delta L = 1 + \lambda_N L$. Initially its right-hand side is in L^2 , so boundary regularity gives $L \in H^2(\Omega)$. Repeated application of higher-order boundary regularity gives $L \in H^k(\Omega)$ for every integer k , and Sobolev embedding then gives $L \in C^\infty(\overline{\Omega})$; see [1, Theorem 4.31 and Corollary 4.32, p. 116].

Define $L^- = \max\{-L, 0\} = (|L| - L)/2$. By Fact 2.1, $|L| \in H_0^1(\Omega)$ because $L \in H_0^1(\Omega)$; linearity of this space therefore gives $L^- \in H_0^1(\Omega)$. Almost everywhere on $\{L < 0\}$, $L = -L^-$ and $\nabla L = -\nabla L^-$; on its complement L^- and ∇L^- vanish almost everywhere. Taking $\eta = L^-$ in equation (2.4) therefore gives

$$-b(L^-, L^-) = b(L, L^-) = \int_{\Omega} L^- \geq 0.$$

Coercivity forces $L^- = 0$, and thus $L \geq 0$ everywhere by continuity. Consequently

$$-\Delta L = 1 + \lambda_N L > 0 \quad \text{in } \Omega.$$

In fact $L > 0$ in Ω : an interior zero would be a local minimum of the smooth nonnegative function L , where $\Delta L \geq 0$, contrary to the displayed equation. Apply Fact 2.5 to $u = -L$, which satisfies

$u < 0$ inside, $u = 0$ on the boundary, and $\Delta u = 1 + \lambda_N L > 0$. The smooth boundary satisfies the interior sphere condition, so

$$-\partial_\nu L > 0 \quad \text{on } \partial\Omega, \quad (2.5)$$

where ν is the outward unit normal; see [1, Lemma 2.21, p. 29].

For $v \in V$, the equations $-\Delta v = \lambda_N v$ and $(-\Delta - \lambda_N)L = 1$, followed by Green's identity, give

$$\begin{aligned} 0 &= \int_{\Omega} v = \int_{\Omega} (v(-\Delta L) - L(-\Delta v)) \\ &= \int_{\partial\Omega} (L\partial_\nu v - v\partial_\nu L) d\sigma \\ &= - \int_{\partial\Omega} v\partial_\nu L d\sigma. \end{aligned}$$

All functions are smooth up to the boundary, so this is the classical Green identity. In the last line we used $L = 0$ and $\partial_\nu v = 0$ on $\partial\Omega$. The boundary diffeomorphism $\Phi : S^1 \rightarrow \partial\Omega$ changes arclength by $d\sigma = |\Phi'(s)| ds$. Hence

$$0 = \int_{S^1} \beta(s) h_v(s) ds, \quad \beta(s) = |\Phi'(s)| (-\partial_\nu L)(\Phi(s)).$$

This β depends only on Ω and the fixed parametrization Φ , and is smooth and strictly positive by equation (2.5) and lemma 2.2.

If h_v were identically zero, boundary surjectivity of Φ would give $\text{Tr } v = 0$, hence $v \in H_0^1(\Omega)$. Testing equation (2.2) with $\psi = v$ and applying equation (2.3) would give

$$\lambda_D \int_{\Omega} v^2 \leq \int_{\Omega} |\nabla v|^2 = \lambda_N \int_{\Omega} v^2.$$

The strict spectral gap forces $v = 0$. Thus $v \neq 0$ implies that h_v is a nonzero continuous function. If it were everywhere nonnegative, strict positivity of β and continuity would give $\int_{S^1} \beta h_v ds > 0$; if it were everywhere nonpositive, the integral would be negative. Both contradict its zero weighted integral. Therefore h_v takes both signs. $\square$

Fact 2.6 (Restrictions to sign components). Let $v \in C^1(\Omega) \cap H^1(\Omega; \mathbb{R})$, and let U be a connected component of $\{v > 0\}$ or $\{v < 0\}$. Define $v_U = v \mathbf{1}_U$. Then $v_U \in H^1(\Omega)$ and

$$\nabla v_U = \mathbf{1}_U \nabla v \quad \text{almost everywhere in } \Omega. \quad (2.6)$$

Proof. Choose a ball B with $\overline{B} \subset \Omega$. The function v is Lipschitz on B , with constant $M = \sup_B |\nabla v|$. If $x, y \in B$ both belong to U , the same Lipschitz estimate holds for v_U ; if neither belongs to U , their v_U values are both zero. If $x \in U$ and $y \notin U$, the segment from x to y first exits U at a point $z \in \partial U \cap B$. One has $v(z) = 0$: continuity excludes the opposite sign, and a nonzero value of the same sign would give a small connected neighbourhood meeting U and lying in the same sign set, contradicting $z \notin U$. Thus

$$|v_U(x) - v_U(y)| = |v(x) - v(z)| \leq M|x - z| \leq M|x - y|.$$

The remaining mixed case follows by exchanging x and y . Hence v_U is locally Lipschitz and has local weak derivatives. On U its gradient equals ∇v . At any point outside U where $v \neq 0$, it is locally zero: a sufficiently small connected ball is contained in one sign component. On $\{v = 0\}$,

the gradient of v_U is zero almost everywhere by the zero-level-set property in Fact 2.1, applied locally to v_U . This proves equation (2.6). Since $|v_U| \leq |v|$ and $|\nabla v_U| \leq |\nabla v|$ almost everywhere, the function and its local weak derivatives belong globally to L^2 . The local distributional identities therefore give $v_U \in H^1(\Omega)$. $\square$

Fact 2.7 (Interior uniqueness for the Helmholtz equation). If $W \subset \mathbb{R}^2$ is connected and open, $\lambda \geq 0$, and $f \in C^2(W)$ satisfies $-\Delta f = \lambda f$, then f is real analytic. If f vanishes on a nonempty open subset of W , it vanishes throughout W .

Proof. The function $F(x, t) = e^{\sqrt{\lambda}t} f(x)$ is harmonic on $W \times \mathbb{R}$, because $\Delta_{x,t} F = e^{\sqrt{\lambda}t} (\Delta_x f + \lambda f) = 0$. Harmonic functions are real analytic [1, Theorem 2.10, p. 24], so restriction to $t = 0$ makes f real analytic. For completeness, let Z be the set of points of W where every derivative of f vanishes. This set is relatively closed by continuity of all derivatives. At any point of Z , the convergent Taylor series makes f zero in a neighbourhood, so Z is also open. If f vanishes on a nonempty open set, Z is nonempty; connectedness then gives $Z = W$. $\square$

Fact 2.8 (Separation by a disk crosscut). A crosscut of $\mathbb{D}$ is a simple arc α , the image of a continuous injective map $[0, 1] \rightarrow \overline{\mathbb{D}}$, whose distinct endpoints s_1, s_3 lie on S^1 and whose remaining points lie in $\mathbb{D}$. If I_1, I_2 are the two closed circle arcs between those endpoints, then $\mathbb{D} \setminus \alpha$ has exactly two connected components C_1, C_2 , numbered so that

$$\partial C_j = \alpha \cup I_j \quad (j = 1, 2).$$

Consequently, if s_1, s_2, s_3, s_4 are cyclically ordered on S^1 , every continuous path from s_2 to s_4 whose remaining points lie in $\mathbb{D}$ intersects α inside $\mathbb{D}$.

The separation statement is the crosscut form of the Jordan curve theorem; see [5, Section 14.2, “Crosscuts”]. To justify the consequence explicitly, number the circle arcs so that $s_2 \in I_1 \setminus \{s_1, s_3\}$ and $s_4 \in I_2 \setminus \{s_1, s_3\}$. Then $s_2 \notin \overline{C_2}$ and $s_4 \notin \overline{C_1}$. If the path avoided α , its interior points near s_2 would belong to C_1 and those near s_4 to C_2 . Its connected interior image cannot meet both components while remaining in $\mathbb{D} \setminus \alpha$.

Lemma 2.4. *For every $0 \neq v \in V$, the sets $\{v > 0\}$ and $\{v < 0\}$ in Ω are connected. The boundary trace h_v cannot have four cyclically ordered points with strictly alternating signs.*

Proof. Both sign sets are nonempty: v is continuous, nonzero and has mean zero. For any component U of either sign set, Fact 2.6 gives $v_U \in H^1(\Omega)$ and its weak gradient. We may therefore take $\psi = v_U$ in equation (2.2), obtaining

$$\int_{\Omega} |\nabla v_U|^2 = \lambda_N \int_{\Omega} v_U^2. \quad (2.7)$$

Suppose there were at least three distinct sign components U_1, U_2, U_3 . Set $A_j = \int_{\Omega} v_{U_j}$ for $j = 1, 2$. Each A_j is nonzero, because v_{U_j} has one strict sign on the nonempty open component U_j and is zero elsewhere. The function

$$f = A_2 v_{U_1} - A_1 v_{U_2}$$

belongs to $H^1(\Omega)$, has mean zero, and is nonzero, for example on U_1 . The two restrictions and their gradients vanish almost everywhere outside the respective disjoint sets U_1, U_2 , by equation (2.6); hence all cross terms in their energies are zero. Thus equation (2.7) gives

$$\int_{\Omega} |\nabla f|^2 = \lambda_N \int_{\Omega} f^2.$$

By Fact 2.2, f is a weak Neumann eigenfunction. Elliptic regularity makes it smooth in the interior. It vanishes on U_3 , so Fact 2.7 forces $f = 0$ throughout the connected domain Ω , a contradiction. There are therefore exactly two sign components, one positive and one negative.

For the boundary assertion, put $\hat{v} = v \circ \Phi$ on $\mathbb{D}$. The homeomorphism Φ makes the interior sets $P = \{\hat{v} > 0\} \cap \mathbb{D}$ and $N = \{\hat{v} < 0\} \cap \mathbb{D}$ connected and open. Suppose $s_1, s_2, s_3, s_4 \in S^1$ are distinct and cyclically ordered, with positive values at s_1, s_3 and negative values at s_2, s_4 ; the other alternating pattern is handled by replacing v by $-v$. Continuity gives short inward radial segments at s_1 and s_3 whose values are strictly positive, including their boundary endpoints. Their interior endpoints can be joined by a finite polygonal path in P . Indeed, an open connected subset of $\mathbb{R}^2$ is polygonally connected: the points reachable from a fixed point by finite polygonal paths form a relatively open set with relatively open complement. The union of this path and the two radial segments contains a simple polygonal arc α from s_1 to s_3 . To obtain this arc, subdivide the finitely many segments at their intersections and at endpoints of overlaps. The resulting finite connected embedded graph has a simple graph path between s_1 and s_3 . Its image is the required arc, whose other points lie in P .

Construct in the same way a continuous path γ from s_2 to s_4 , with every other point in N , using negative radial segments and a polygonal path in N . By Fact 2.8, γ meets α inside $\mathbb{D}$. At the intersection $\hat{v}$ would be both strictly positive and strictly negative. This contradiction proves the boundary assertion. Figure 1 illustrates the configuration; the separation argument does not impose any straightness or curvature condition on the connecting paths. $\square$

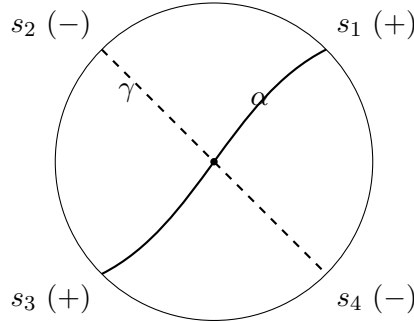


Figure 1: Schematic of the forbidden alternating-sign configuration. The positive crosscut α joins s_1 to s_3 . The putative negative path γ joins the other two boundary points and must intersect α by Fact 2.8. The particular shapes drawn are illustrative; the theorem applies to the paths constructed in lemma 2.4.

The following bound is due to Nadirashvili [6, Theorem 3, p. 227]; see also [7, Proposition 2.5]. We give the boundary-sign proof.

Proposition 2.5. *The real eigenspace V has dimension at most two.*

Proof. Suppose $\dim_{\mathbb{R}} V \geq 3$. Consider the real linear map

$$T : V \rightarrow \mathbb{R}^2, \quad T(v) = \left(\int_{S^1} \beta h_v \cos \theta \, ds, \int_{S^1} \beta h_v \sin \theta \, ds \right), \quad s = e^{i\theta}.$$

Its rank is at most two, so its kernel contains a nonzero v . By lemma 2.3, the trace $h = h_v$ is nonzero, takes both signs, and satisfies

$$\int_{S^1} \beta h \, ds = \int_{S^1} \beta h \cos \theta \, ds = \int_{S^1} \beta h \sin \theta \, ds = 0. \quad (2.8)$$

Choose a strictly negative point $e^{i\theta_0}$ and cut the circle there, parametrizing the other points by $\theta \in (\theta_0, \theta_0 + 2\pi)$. Write $H(\theta) = h(e^{i\theta})$, $B(\theta) = \beta(e^{i\theta})$, and

$$A = \{\theta \in (\theta_0, \theta_0 + 2\pi) : H(\theta) > 0\}, \quad a = \inf A, \quad b = \sup A.$$

The set A is nonempty and open. Continuity and strict negativity at the cut give negative neighbourhoods at both ends of the angle interval. Therefore

$$\theta_0 < a < b < \theta_0 + 2\pi.$$

Continuity gives $H(a) = H(b) = 0$, and the definitions give $H \leq 0$ outside $[a, b]$. There is no negative value at any $\theta \in (a, b)$: the definitions of infimum and supremum provide positive angles $\theta_- < \theta < \theta_+$ in A . The cut point, $e^{i\theta_-}$, $e^{i\theta}$ and $e^{i\theta_+}$ would then have strictly alternating signs in cyclic order, contradicting lemma 2.4. Hence $H \geq 0$ on $[a, b]$.

Set $c = (a + b)/2$ and $d = (b - a)/2 \in (0, \pi)$, and define the 2π -periodic function

$$F(\theta) = \cos(\theta - c) - \cos d.$$

On the circle, $F > 0$ precisely on the arc represented by (a, b) , $F = 0$ at its endpoints, and $F < 0$ on the complementary open arc. This follows from strict decrease of cosine on $[0, \pi]$, using angular distance from c in $[0, \pi]$. Thus $FH \geq 0$ everywhere, and $FH > 0$ on the nonempty open set A . Since $B > 0$,

$$\int_{\theta_0}^{\theta_0+2\pi} B(\theta)F(\theta)H(\theta) d\theta > 0.$$

On the other hand,

$$F(\theta) = \cos c \cos \theta + \sin c \sin \theta - \cos d,$$

so equation (2.8), together with $ds = d\theta$, makes this same integral zero. The contradiction proves $\dim_{\mathbb{R}} V \leq 2$. $\square$

3 A variational principle for tangent vector fields

For $X \in H^1(\Omega; \mathbb{R}^2)$, *tangent trace* means $\text{Tr } X \cdot \nu = 0$ almost everywhere on $\partial\Omega$; see [9, Theorem 1.1, p. 2] for this boundary condition. For any $X \in H^1(\Omega; \mathbb{R}^2)$ define

$$Q(X) = \int_{\Omega} (|\text{div } X|^2 + |\text{curl } X|^2), \quad \text{curl } X = \partial_1 X_2 - \partial_2 X_1.$$

The following is the divergence–curl form of Rohleder’s vector variational principle [9, Theorem 1.1]; see also [34]. We give its derivation, including the equality case.

Proposition 3.1. *For every $X \in H^1(\Omega; \mathbb{R}^2)$ with tangent trace,*

$$Q(X) \geq \lambda_N \|X\|_2^2. \tag{3.1}$$

Equality holds if and only if $X = \nabla v$ for some $v \in V$.

Proof. We use the Sobolev integration-by-parts identity

$$\int_{\Omega} Y \cdot \nabla u = - \int_{\Omega} u \text{div } Y + \int_{\partial\Omega} \text{Tr } u (\text{Tr } Y \cdot \nu) d\sigma, \tag{3.2}$$

valid for $u \in H^1(\Omega; \mathbb{R})$ and $Y \in H^1(\Omega; \mathbb{R}^2)$. It follows from the classical divergence theorem by smooth approximation in H^1 and continuity of the trace into $L^2(\partial\Omega)$; see [1, Theorem 1.46, Theorem 3.44, and Section 3.11]. The boundary product is integrable by Cauchy–Schwarz.

First solve

$$\begin{aligned} \Delta v &= \operatorname{div} X, & \partial_\nu v &= 0, & \int_\Omega v &= 0, \\ \Delta j &= \operatorname{curl} X, & \operatorname{Tr} j &= 0. \end{aligned} \tag{3.3}$$

For the Neumann problem, the required compatibility condition is

$$\int_\Omega \operatorname{div} X = \int_{\partial\Omega} \operatorname{Tr} X \cdot \nu \, d\sigma = 0.$$

Here we applied equation (3.2) with $u = 1$ and $Y = X$. On the closed mean-zero subspace

$$H = \left\{ w \in H^1(\Omega; \mathbb{R}) : \int_\Omega w = 0 \right\},$$

the form $a(v, w) = \int_\Omega \nabla v \cdot \nabla w$ is continuous and coercive by the mean-zero Poincaré inequality. Since $\operatorname{div} X \in L^2(\Omega)$, Lax–Milgram [1, Theorem 4.20, p. 103] gives a unique $v \in H$ such that

$$\int_\Omega \nabla v \cdot \nabla w = - \int_\Omega (\operatorname{div} X) w \quad (w \in H).$$

Compatibility extends this identity to all $w \in H^1(\Omega; \mathbb{R})$ by subtracting their means. It is the weak formulation of the Neumann problem in equation (3.3). Similarly, the Dirichlet Poincaré inequality makes a coercive on $H_0^1(\Omega; \mathbb{R})$, so Lax–Milgram gives a unique $j \in H_0^1(\Omega; \mathbb{R})$ satisfying

$$\int_\Omega \nabla j \cdot \nabla w = - \int_\Omega (\operatorname{curl} X) w \quad (w \in H_0^1(\Omega; \mathbb{R})).$$

No compatibility condition is needed for this Dirichlet problem. Boundary regularity gives $v, j \in H^2(\Omega)$; see [8, Chapter III, Theorems 6.1 and 6.4].

Put $\nabla^\perp j = (-\partial_2 j, \partial_1 j)$. Tangential differentiation commutes with taking the trace for H^2 functions on a smooth domain; this follows by smooth approximation and continuity of the trace maps. Thus the zero trace of j implies that $\nabla^\perp j$ has tangent trace. Consequently

$$Z = X - \nabla v - \nabla^\perp j$$

belongs to $H^1(\Omega; \mathbb{R}^2)$ and has tangent trace. The equations in equation (3.3) give $\operatorname{div} Z = \operatorname{curl} Z = 0$. Its components are harmonic in distributions and hence smooth in the interior by Weyl’s lemma. Simple connectivity therefore gives a smooth real interior potential r with $Z = \nabla r$.

For $M > 0$, define $r_M = \max\{-M, \min\{r, M\}\}$, and write $s^+ = \max\{s, 0\}$. The weak derivative formulas for positive parts [1, Proposition 3.22 and the subsequent discussion, pp. 57–58], applied locally to $r_M = r - (r - M)^+ + (-r - M)^+$, give

$$r_M \in H_{\operatorname{loc}}^1(\Omega), \quad \nabla r_M = \mathbf{1}_{\{|r| < M\}} Z \quad \text{almost everywhere.}$$

Since $|r_M| \leq M$, Ω is bounded, and $Z \in L^2(\Omega)$, it follows that $r_M \in H^1(\Omega)$.

Fix a ball B with $\overline{B} \subset \Omega$, and write u_B and u_Ω for the averages over B and Ω . The usual mean-zero Poincaré inequality gives, for $u \in H^1(\Omega)$,

$$\begin{aligned} |u_B - u_\Omega| &\leq |B|^{-1/2} \|u - u_\Omega\|_2, \\ \|u - u_B\|_2 &\leq \left(1 + \sqrt{\frac{|\Omega|}{|B|}}\right) \|u - u_\Omega\|_2 \\ &\leq C_{\Omega,B} \|\nabla u\|_2. \end{aligned}$$

Applying this to r_M and using $|r_M| \leq |r|$ on B yields

$$\|r_M\|_2 \leq C_{\Omega,B} \|Z\|_2 + |\Omega|^{1/2} |(r_M)_B| \leq C_{\Omega,B} \|Z\|_2 + |\Omega|^{1/2} \sup_B |r|.$$

The bound is independent of M . Since $r_M \rightarrow r$ pointwise, Fatou's lemma gives $r \in L^2(\Omega)$. Moreover, for every $\phi \in C_c^\infty(\Omega)$,

$$\int_\Omega r \partial_i \phi = - \int_\Omega Z_i \phi \quad (i = 1, 2),$$

by the interior identity $\nabla r = Z$. Thus the global weak gradient of r is Z , and $r \in H^1(\Omega)$. This establishes its global integrability using only the interior potential and $Z \in L^2$. Applying equation (3.2) with $u = r$ and $Y = Z$ now gives

$$\int_\Omega |\nabla r|^2 = \int_\Omega Z \cdot \nabla r = - \int_\Omega r \operatorname{div} Z + \int_{\partial\Omega} \operatorname{Tr} r (\operatorname{Tr} Z \cdot \nu) d\sigma = 0.$$

Hence $Z = 0$, and

$$X = \nabla v + \nabla^\perp j. \tag{3.4}$$

The summands are orthogonal in L^2 , since

$$\int_\Omega \nabla v \cdot \nabla^\perp j = - \int_\Omega v \operatorname{div}(\nabla^\perp j) + \int_{\partial\Omega} \operatorname{Tr} v (\operatorname{Tr} \nabla^\perp j \cdot \nu) d\sigma = 0.$$

Consequently

$$\|X\|_2^2 = \|\nabla v\|_2^2 + \|\nabla j\|_2^2, \quad Q(X) = \|\Delta v\|_2^2 + \|\Delta j\|_2^2.$$

Integration by parts, Cauchy-Schwarz, and the two Poincaré inequalities give

$$\begin{aligned} \|\nabla v\|_2^2 &= - \int_\Omega v \Delta v \leq \|v\|_2 \|\Delta v\|_2 \\ &\leq \lambda_N^{-1/2} \|\nabla v\|_2 \|\Delta v\|_2, \\ \|\nabla j\|_2^2 &= - \int_\Omega j \Delta j \leq \|j\|_2 \|\Delta j\|_2 \\ &\leq \lambda_D^{-1/2} \|\nabla j\|_2 \|\Delta j\|_2. \end{aligned}$$

The boundary terms vanish because $\partial_\nu v = 0$ and $\operatorname{Tr} j = 0$, respectively. The last inequalities use the zero mean of v and the zero trace of j . Dividing by the gradient norms when nonzero, with the zero cases immediate, gives

$$\|\Delta v\|_2^2 \geq \lambda_N \|\nabla v\|_2^2, \quad \|\Delta j\|_2^2 \geq \lambda_D \|\nabla j\|_2^2.$$

It follows that

$$\begin{aligned} Q(X) &\geq \lambda_N \|\nabla v\|_2^2 + \lambda_D \|\nabla j\|_2^2 \\ &= \lambda_N \|X\|_2^2 + (\lambda_D - \lambda_N) \|\nabla j\|_2^2. \end{aligned} \quad (3.5)$$

This proves equation (3.1).

Suppose equality holds in equation (3.1). Since $\lambda_D > \lambda_N$, equation (3.5) forces $\nabla j = 0$, and the zero trace gives $j = 0$. Thus $X = \nabla v$ and $\|\Delta v\|_2^2 = \lambda_N \|\nabla v\|_2^2$. If $\nabla v \neq 0$, the preceding chain becomes

$$\begin{aligned} \|\nabla v\|_2^2 &\leq \|v\|_2 \|\Delta v\|_2 \\ &\leq \lambda_N^{-1/2} \|\nabla v\|_2 \|\Delta v\|_2 = \|\nabla v\|_2^2. \end{aligned}$$

Both inequalities are equalities, so $\|\nabla v\|_2^2 = \lambda_N \|v\|_2^2$. The equality characterization of the mean-zero Neumann variational inequality (Fact 2.2 of the spectral and planar preliminaries) gives equation (2.2), hence $v \in V$. If $\nabla v = 0$, its zero mean gives $v = 0 \in V$.

Conversely, if $v \in V$, then $X = \nabla v$ belongs to $H^1(\Omega; \mathbb{R}^2)$ and has tangent trace because $\partial_\nu v = 0$. Also $\text{curl } X = 0$ and $\text{div } X = \Delta v = -\lambda_N v$, so

$$Q(X) = \lambda_N^2 \|v\|_2^2 = \lambda_N \|\nabla v\|_2^2 = \lambda_N \|X\|_2^2,$$

where the middle equality follows by testing equation (2.2) with v . $\square$

For a differential-form interpretation of the divergence–curl energy, see [22, Section 5.1].

We next record the boundary identity in fixed Cartesian coordinates. Let $\tau = (-\nu_2, \nu_1)$ be the counterclockwise unit tangent, and set

$$\kappa = \det(\tau, \partial_\tau \tau).$$

Thus κ is positive on a convex circle. Here $|\nabla X|^2 = \sum_{i,j=1}^2 (\partial_i X_j)^2$.

Lemma 3.2. *For every $X \in H^1(\Omega; \mathbb{R}^2)$ with tangent trace,*

$$Q(X) = \int_\Omega |\nabla X|^2 + \int_{\partial\Omega} \kappa |\text{Tr } X|^2 d\sigma. \quad (3.6)$$

Proof. First suppose $X \in C^\infty(\overline{\Omega}; \mathbb{R}^2)$, without imposing tangency. Expanding the bulk integrands gives

$$\begin{aligned} &|\text{div } X|^2 + |\text{curl } X|^2 - |\nabla X|^2 \\ &= (\partial_1 X_1 + \partial_2 X_2)^2 + (\partial_1 X_2 - \partial_2 X_1)^2 \\ &\quad - \sum_{i,j=1}^2 (\partial_i X_j)^2 \\ &= 2(\partial_1 X_1 \partial_2 X_2 - \partial_1 X_2 \partial_2 X_1) = 2 \det \nabla X. \end{aligned}$$

Put

$$F = (X_1 \partial_2 X_2 - X_2 \partial_2 X_1, X_2 \partial_1 X_1 - X_1 \partial_1 X_2).$$

The mixed second derivatives cancel, so $\text{div } F = 2 \det \nabla X$. On the boundary,

$$F \cdot \nu = X_1 \partial_\tau X_2 - X_2 \partial_\tau X_1.$$

The divergence theorem therefore gives

$$Q(X) - \int_{\Omega} |\nabla X|^2 = \int_{\partial\Omega} (X_1 \partial_{\tau} X_2 - X_2 \partial_{\tau} X_1) d\sigma. \quad (3.7)$$

For $h = (h_1, h_2) \in H^{1/2}(\partial\Omega; \mathbb{R}^2)$, define

$$B(h) = \langle \partial_{\tau} h_2, h_1 \rangle - \langle \partial_{\tau} h_1, h_2 \rangle,$$

where the brackets denote the real $H^{-1/2}$ - $H^{1/2}$ duality pairing, agreeing with the boundary integral for smooth functions. Tangential differentiation is bounded from $H^{1/2}(\partial\Omega)$ to $H^{-1/2}(\partial\Omega)$, so B is continuous on $H^{1/2}$. Choose $X_n \in C^{\infty}(\bar{\Omega}; \mathbb{R}^2)$ converging to $X \in H^1(\Omega; \mathbb{R}^2)$ in H^1 . Their traces converge to $h = \text{Tr } X$ in $H^{1/2}$, hence $B(\text{Tr } X_n) \rightarrow B(h)$. The bulk integrals also converge, since the first derivatives converge in L^2 . Passing to the limit in equation (3.7) gives

$$Q(X) - \int_{\Omega} |\nabla X|^2 = B(h). \quad (3.8)$$

The approximating fields need not have tangent trace.

Now suppose h is tangent, and put $a = h \cdot \tau$, so $h = a\tau$. Multiplication by the smooth components of τ preserves $H^{1/2}$, and hence $a \in H^{1/2}(\partial\Omega; \mathbb{R})$. First take a smooth. The product rule gives

$$\begin{aligned} B(a\tau) &= \int_{\partial\Omega} [a\tau_1 \partial_{\tau}(a\tau_2) - a\tau_2 \partial_{\tau}(a\tau_1)] d\sigma \\ &= \int_{\partial\Omega} a^2 \det(\tau, \partial_{\tau}\tau) d\sigma = \int_{\partial\Omega} \kappa a^2 d\sigma, \end{aligned}$$

because the terms containing $a \partial_{\tau} a$ cancel. For general $a \in H^{1/2}$, choose smooth $a_n \rightarrow a$ in $H^{1/2}$. Then $a_n \tau \rightarrow a\tau$ in $H^{1/2}$, so continuity of B gives $B(a_n \tau) \rightarrow B(a\tau)$. Also $a_n \rightarrow a$ in L^2 , and

$$\|a_n^2 - a^2\|_{L^1(\partial\Omega)} \leq \|a_n - a\|_{L^2(\partial\Omega)} (\|a_n\|_{L^2(\partial\Omega)} + \|a\|_{L^2(\partial\Omega)}) \rightarrow 0.$$

Since κ is bounded, we may pass to the limit to obtain $B(a\tau) = \int_{\partial\Omega} \kappa a^2 d\sigma$. Finally $|h|^2 = a^2$, and equation (3.8) proves equation (3.6). $\square$

4 Boundary kernels and the multiplier identity

4.1 A subcritical disk potential

Let $\Omega \subset \mathbb{R}^2$ be a nonempty bounded simply connected open set with smooth boundary, and fix the smooth conformal diffeomorphism $\Phi : \mathbb{D} \rightarrow \bar{\Omega}$ constructed in lemma 2.2. Here $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and $S^1 = \partial\mathbb{D}$. Write λ_D and λ_N for the first Dirichlet and first positive Neumann eigenvalues of Ω , respectively. Define

$$dq(x) = \lambda_N |\Phi'(x)|^2 dx. \quad (4.1)$$

Its density is bounded and positive. We use real scalar functions and write $\langle f, g \rangle_q = \int_{\mathbb{D}} fg dq$ and $\|f\|_q^2 = \langle f, f \rangle_q$.

For $\eta \in H_0^1(\mathbb{D})$, put $u = \eta \circ \Phi^{-1}$. The smooth diffeomorphism preserves zero trace, so $u \in H_0^1(\Omega)$. Conformality gives $D\Phi = |\Phi'|R$, where R is a rotation, and $\det D\Phi = |\Phi'|^2$. Thus the chain rule and change of variables give

$$\int_{\Omega} |\nabla u|^2 dy = \int_{\mathbb{D}} |\nabla \eta|^2 dx.$$

These identities hold first for smooth functions and then by density in H_0^1 . The Dirichlet inequality on Ω therefore gives

$$\|\eta\|_q^2 = \lambda_N \int_{\Omega} u^2 dy \leq \frac{\lambda_N}{\lambda_D} \int_{\Omega} |\nabla u|^2 dy = \frac{\lambda_N}{\lambda_D} \int_{\mathbb{D}} |\nabla \eta|^2 dx.$$

The spectral comparison lemma 2.1 gives $\lambda_N < \lambda_D$. Hence

$$\|\eta\|_q^2 \leq d \int_{\mathbb{D}} |\nabla \eta|^2 dx \quad (\eta \in H_0^1(\mathbb{D})), \quad d = \frac{\lambda_N}{\lambda_D} < 1. \quad (4.2)$$

For the scalar kernel construction, we allow any measure $dq = a dx$ with bounded nonnegative density satisfying equation (4.2) for some fixed $0 < d < 1$. The conformal measure equation (4.1) is one such measure; the zero measure is also allowed.

The Dirichlet Green function of the disk is

$$G(x, y) = \frac{1}{2\pi} \log \left| \frac{1 - \bar{x}y}{x - y} \right| = \frac{1}{4\pi} \log \left(1 + \frac{(1 - |x|^2)(1 - |y|^2)}{|x - y|^2} \right), \quad x \neq y. \quad (4.3)$$

It is nonnegative and symmetric, vanishes when either variable lies on S^1 , and has logarithmic singularity $-(2\pi)^{-1} \log |x - y|$. Its diagonal value is set to $+\infty$; the diagonal has measure zero in all integrals below. For fixed $x \in \mathbb{D}$, it is harmonic in y away from x : both logarithms in the first formula are logarithms of moduli of nonvanishing holomorphic functions there.

Lemma 4.1. *The operator*

$$Tf(x) = \int_{\mathbb{D}} G(x, y) f(y) dq(y)$$

is bounded and self-adjoint on $L^2(q)$, with $\|T\| \leq d < 1$. For each $f \in L^2(q)$, the displayed integral is the continuous interior representative of the weak Dirichlet solution of $-\Delta\psi = af$. For every measure $0 \leq \rho \leq q$, the operator

$$T_\rho f(x) = \int_{\mathbb{D}} G(x, y) f(y) d\rho(y)$$

has the same properties on $L^2(\rho)$, with $\|T_\rho\| \leq d$.

Proof. We first justify the Green representation for a source $b \in L^2(\mathbb{D}; dx)$. The Poincaré inequality and Lax–Milgram [1, Theorem 4.20, p. 103] give a unique $u \in H_0^1(\mathbb{D})$ satisfying

$$\int_{\mathbb{D}} \nabla u \cdot \nabla \eta dx = \int_{\mathbb{D}} b \eta dx \quad (\eta \in H_0^1(\mathbb{D})). \quad (4.4)$$

If $b \in C_c^\infty(\mathbb{D})$, this solution is smooth up to the boundary. Apply Green’s second identity to u and $G(x, \cdot)$ on $\mathbb{D} \setminus \overline{B_\varepsilon(x)}$. The outer boundary term is zero. On the inner circle, whose outward normal points towards x ,

$$G(x, y) = -\frac{1}{2\pi} \log \varepsilon + O(1), \quad \partial_\nu G(x, y) = \frac{1}{2\pi\varepsilon} + O(1).$$

Consequently the integral of $G\partial_\nu u - u\partial_\nu G$ tends to $-u(x)$. Since $\Delta_y G(x, y) = 0$ away from x and $\Delta u = -b$, letting $\varepsilon \downarrow 0$ gives

$$u(x) = \int_{\mathbb{D}} G(x, y) b(y) dy. \quad (4.5)$$

For general $b \in L^2(dx)$, choose $b_n \in C_c^\infty(\mathbb{D})$ converging to b in L^2 , and let u_n be their weak solutions. Testing the difference equation and using Poincaré gives

$$\|\nabla(u_n - u)\|_2^2 \leq C\|b_n - b\|_2\|\nabla(u_n - u)\|_2.$$

Thus $u_n \rightarrow u$ in H_0^1 , and a subsequence converges almost everywhere. For each fixed $x \in \mathbb{D}$, the logarithmic pole is square integrable, so $G(x, \cdot) \in L^2(dx)$ and

$$\left| \int_{\mathbb{D}} G(x, y)(b_n - b)(y) dy \right| \leq \|G(x, \cdot)\|_2 \|b_n - b\|_2 \rightarrow 0.$$

Passing to the limit in equation (4.5) proves that its right-hand side equals u almost everywhere.

To prove continuity of this representative, fix $K \Subset \mathbb{D}$ and $r < 1$ with $|x| \leq r$ on K . For $x, x_0 \in K$,

$$|\log|1 - \bar{x}y| - \log|1 - \bar{x}_0y|| \leq \frac{|x - x_0|}{1 - r} \quad (y \in \mathbb{D}).$$

For the singular part choose $\chi \in C_c^\infty(\mathbb{R}^2)$ equal to one on $\{|z| \leq 2\}$ and set $F(z) = \chi(z) \log|z|$. Then $F \in L^2(\mathbb{R}^2)$, since $\int_0^1 t |\log t|^2 dt < \infty$, and

$$\|\log|x - \cdot| - \log|x_0 - \cdot|\|_{L^2(\mathbb{D})} \leq \|F(x - \cdot) - F(x_0 - \cdot)\|_{L^2(\mathbb{R}^2)} \rightarrow 0.$$

Here continuity of translations in L^2 follows by approximation by smooth compactly supported functions: translation preserves the L^2 norm, and the translated approximants converge in L^2 . Together with equation (4.3), these estimates give $\|G(x, \cdot) - G(x_0, \cdot)\|_2 \rightarrow 0$. Hence

$$\left| \int_{\mathbb{D}} (G(x, y) - G(x_0, y))b(y) dy \right| \leq \|G(x, \cdot) - G(x_0, \cdot)\|_2 \|b\|_2 \rightarrow 0.$$

This proves the asserted interior continuity.

Now let $f \in L^2(q)$. The source af belongs to $L^2(dx)$ because

$$\int_{\mathbb{D}} |af|^2 dx \leq \|a\|_\infty \int_{\mathbb{D}} |f|^2 dq.$$

Changing f on a q -null set does not change af almost everywhere with respect to dx , so the integral defines a linear map on $L^2(q)$. The representation just proved identifies $\psi = Tf$ with the weak solution for $b = af$. Taking $\eta = \psi$ in equation (4.4) gives

$$\int_{\mathbb{D}} |\nabla\psi|^2 dx = \int_{\mathbb{D}} af\psi dx = \langle \psi, f \rangle_q.$$

Thus equation (4.2) and Cauchy-Schwarz imply

$$\|\psi\|_q^2 \leq d \int_{\mathbb{D}} |\nabla\psi|^2 dx = d \langle \psi, f \rangle_q \leq d \|\psi\|_q \|f\|_q.$$

Dividing when $\|\psi\|_q > 0$, with the zero case immediate, proves $\|T\| \leq d$.

For self-adjointness, equation (4.3) gives $G(x, y) = G(y, x)$ and

$$0 \leq G(x, y) \leq \frac{1}{2\pi} \log \frac{2}{|x - y|}.$$

The square of the upper bound is integrable on $\mathbb{D} \times \mathbb{D}$; boundedness of a therefore gives $G \in L^2(q \times q)$. For $f, g \in L^2(q)$, Cauchy–Schwarz on the product space gives

$$\iint G(x, y) |f(y)g(x)| \, dq(y) \, dq(x) \leq \|G\|_{L^2(q \times q)} \|f\|_q \|g\|_q < \infty.$$

Fubini and symmetry now yield

$$\langle Tf, g \rangle_q = \iint G(x, y) f(y)g(x) \, dq(y) \, dq(x) = \langle f, Tg \rangle_q.$$

Since T is bounded and defined on all of $L^2(q)$, it is self-adjoint.

Finally, $0 \leq \rho \leq q$ means $\rho(E) \leq q(E)$ for every measurable set E . It follows that $d\rho = c \, dx$ with $0 \leq c \leq a$ almost everywhere, so c is bounded. Moreover,

$$\|\eta\|_{L^2(\rho)}^2 \leq \|\eta\|_q^2 \leq d \int_{\mathbb{D}} |\nabla \eta|^2 \, dx \quad (\eta \in H_0^1(\mathbb{D})).$$

This is the estimate needed to bound the solution operator on $L^2(\rho)$. Indeed, for $f \in L^2(\rho)$ the source cf lies in $L^2(dx)$, and its weak solution $\psi = T_\rho f$ satisfies

$$\|\psi\|_{L^2(\rho)}^2 \leq d \int_{\mathbb{D}} |\nabla \psi|^2 \, dx = d \int_{\mathbb{D}} f\psi \, d\rho \leq d \|f\|_{L^2(\rho)} \|\psi\|_{L^2(\rho)}.$$

Thus $\|T_\rho\| \leq d$. The same Green representation proves interior continuity, and the same Fubini argument with c in place of a proves self-adjointness. $\square$

For Lipschitz boundary data $h : S^1 \rightarrow \mathbb{R}$, define the Poisson extension

$$Ah(x) = \int_{S^1} P_s(x) h(s) \, ds, \quad P_s(x) = \frac{1 - |x|^2}{2\pi |s - x|^2}, \quad (4.6)$$

where ds is arclength on the unit circle. To justify its Sobolev regularity, write

$$h(e^{i\theta}) = \sum_{n \in \mathbb{Z}} h_n e^{in\theta}, \quad Ah(re^{i\theta}) = \sum_{n \in \mathbb{Z}} h_n r^{|n|} e^{in\theta}.$$

Lipschitz regularity gives $h \in H^1(S^1)$, so $\sum_n (1 + n^2) |h_n|^2 < \infty$. Parseval's identity and integration in polar coordinates give

$$\|Ah\|_{L^2(\mathbb{D})}^2 = \pi \sum_{n \in \mathbb{Z}} \frac{|h_n|^2}{|n| + 1}, \quad \|\nabla Ah\|_{L^2(\mathbb{D})}^2 = 2\pi \sum_{n \in \mathbb{Z}} |n| |h_n|^2.$$

These identities hold first for finite Fourier sums; their harmonic extensions converge in $H^1(\mathbb{D})$ by the displayed bounds. Their boundary values converge to h in $H^1(S^1)$, so continuity of the trace shows that $\text{Tr } Ah = h$. Harmonicity also passes to the limit, giving

$$\int_{\mathbb{D}} \nabla Ah \cdot \nabla \eta \, dx = 0 \quad (\eta \in H_0^1(\mathbb{D})).$$

We seek $W_h \in H^1(\mathbb{D})$ with trace h satisfying

$$\int_{\mathbb{D}} \nabla W_h \cdot \nabla \eta \, dx = \int_{\mathbb{D}} a W_h \eta \, dx \quad (\eta \in H_0^1(\mathbb{D})). \quad (4.7)$$

This is the weak meaning of $-\Delta W_h dx = W_h dq$. Existence and uniqueness follow from Lax–Milgram [1, Theorem 4.20, p. 103] after writing $W_h = Ah + c$, $c \in H_0^1(\mathbb{D})$. Indeed, the continuous form

$$B(u, \eta) = \int_{\mathbb{D}} \nabla u \cdot \nabla \eta dx - \int_{\mathbb{D}} au\eta dx$$

satisfies $B(c, c) \geq (1 - d)\|\nabla c\|_2^2$ by equation (4.2), and the correction equation is $B(c, \eta) = \int_{\mathbb{D}} aAh\eta dx$.

The solution has the explicit representation

$$W_h = Ah + T(I - T)^{-1}Ah. \quad (4.8)$$

To verify it, boundedness of a gives $Ah \in L^2(q)$, and lemma 4.1 gives $(I - T)^{-1} = \sum_{n \geq 0} T^n$ in operator norm on $L^2(q)$. Put $v = (I - T)^{-1}Ah$ and $c = Tv$. Then $c \in H_0^1(\mathbb{D})$, and the weak Poisson equation gives

$$\int_{\mathbb{D}} \nabla c \cdot \nabla \eta dx = \int_{\mathbb{D}} av\eta dx \quad (\eta \in H_0^1(\mathbb{D})).$$

The identity $(I - T)v = Ah$ implies $v = Ah + c = W_h$ in $L^2(q)$. Consequently harmonicity of Ah yields

$$\int_{\mathbb{D}} \nabla W_h \cdot \nabla \eta dx = \int_{\mathbb{D}} \nabla c \cdot \nabla \eta dx = \int_{\mathbb{D}} aW_h\eta dx.$$

Also $\text{Tr } W_h = h$, since c has zero trace. For a direct uniqueness check, the difference z of two solutions with this trace lies in $H_0^1(\mathbb{D})$. Testing its equation by z and using equation (4.2) gives

$$\|\nabla z\|_2^2 = \|z\|_q^2 \leq d\|\nabla z\|_2^2.$$

Since $d < 1$, $\nabla z = 0$, and Poincaré gives $z = 0$. For vector boundary data all constructions and equations are understood componentwise.

4.2 Positive kernels and their estimates

For $s \in S^1$, define $U_{0,s} = P_s$ and recursively set

$$U_{n+1,s}(x) = \int_{\mathbb{D}} G(x, y)U_{n,s}(y) dq(y).$$

The nonnegative integrands define measurable functions with values in $[0, \infty]$. We write $T^n P_s = U_{n,s}$; this definition does not require $P_s \in L^2(q)$. Finiteness is established below. For $p \in \mathbb{D}$ and distinct $s, t \in S^1$, define

$$K(p, s) = P_s(p) + \sum_{n \geq 1} (T^n P_s)(p), \quad (4.9)$$

$$R(s, t) = \sum_{n \geq 0} \int_{\mathbb{D}} P_s T^n P_t dq, \quad (4.10)$$

$$N(s, t) = N_0(s, t) + R(s, t), \quad N_0(s, t) = \frac{1}{\pi|s - t|^2}. \quad (4.11)$$

For integrations we also define R on the diagonal by the same nonnegative series, allowing the value $+\infty$.

Lemma 4.2. *The kernels in equations (4.9) to (4.11) are finite at the stated points. For fixed $p \in \mathbb{D}$, $K(p, \cdot)$ is positive and bounded. The kernel R is symmetric, nonnegative, and integrable on $S^1 \times S^1$, with*

$$\iint_{S^1 \times S^1} R(s, t) ds dt \leq \frac{q(\mathbb{D})}{1-d}. \quad (4.12)$$

For Lipschitz scalar or vector boundary data h, k ,

$$W_h(p) = \int_{S^1} K(p, s) h(s) ds, \quad (4.13)$$

$$\langle Ah, (I - T)^{-1} Ak \rangle_q = \iint_{S^1 \times S^1} R(s, t) h(s) \cdot k(t) ds dt. \quad (4.14)$$

For scalar data the dot denotes multiplication; for vector data the inner product on the left is the sum of the component inner products.

Proof. Constants below are independent of interior and boundary points, unless stated otherwise. The explicit formulas give

$$P_s(y) \leq \frac{1}{\pi|y-s|}, \quad 0 \leq G(x, y) \leq \frac{1}{2\pi} \log \frac{2}{|x-y|}. \quad (4.15)$$

For the first bound use $1 - |y|^2 \leq 2|y-s|$; for the second use $|1 - \bar{x}y| \leq 2$. Translating the centres and integrating over a disk of radius two gives

$$\sup_{s \in S^1} \|P_s\|_{L^{3/2}(dx)} < \infty, \quad \sup_{x \in \mathbb{D}} \|G(x, \cdot)\|_{L^3(dx)} < \infty,$$

since $\int_0^2 r^{-3/2} r dr$ and $\int_0^2 (\log(2/r))^3 r dr$ are finite.

Define the Green potential $F_s = TP_s$. We use it as the starting point for the $L^2(q)$ contraction estimate. Writing $dq = a dy$, Hölder's inequality with exponents 3 and $3/2$ gives

$$F_s(x) = \int_{\mathbb{D}} G(x, y) P_s(y) a(y) dy \leq \|a\|_{\infty} \|G(x, \cdot)\|_{L^3(dx)} \|P_s\|_{L^{3/2}(dx)} \leq C. \quad (4.16)$$

Thus Green integration regularises the singularity of P_s at s , producing a bounded function, with $\|F_s\|_q \leq Cq(\mathbb{D})^{1/2}$. The recursively defined integrals satisfy $T^n P_s = T^{n-1} F_s$ for $n \geq 1$, with the right-hand side interpreted as the $L^2(q)$ operator iterates and, for $n \geq 2$, their Green representatives. Since q has bounded density, the uniform $L^3(dx)$ bound above also gives $\sup_p \|G(p, \cdot)\|_q < \infty$. Hence, for $n \geq 2$,

$$(T^n P_s)(p) = \langle G(p, \cdot), T^{n-2} F_s \rangle_q \leq \|G(p, \cdot)\|_q d^{n-2} \|F_s\|_q \leq C d^{n-2}. \quad (4.17)$$

Together with equation (4.16), this proves convergence of the series for K . For fixed p , the function $P_s(p)$ is positive and bounded as s varies, so the same holds for $K(p, s)$.

For the zeroth term of R , fix $s \neq t$ and put $\delta = |s - t|/3$. On $\mathbb{D} \cap B_{\delta}(s)$, $|y - t| \geq 2\delta$, and equation (4.15) gives

$$P_s(y) P_t(y) \leq \frac{1}{2\pi^2 \delta |y - s|}.$$

This is integrable because $\int_{B_{\delta}(s)} |y - s|^{-1} dy = 2\pi\delta$. The same argument applies near t . Outside these two disks, $P_s P_t \leq (\pi^2 \delta^2)^{-1}$. Thus $P_s P_t \in L^1(dx)$, and boundedness of a gives $\int P_s P_t dq < \infty$.

Also equation (4.15) gives $\sup_s \int_{\mathbb{D}} P_s dq \leq C$. Therefore the explicit bounds equations (4.16) and (4.17) imply

$$\int_{\mathbb{D}} P_s T P_t dq \leq C, \quad \int_{\mathbb{D}} P_s T^n P_t dq \leq C d^{n-2} \quad (n \geq 2).$$

This proves finiteness of $R(s, t)$, and hence of $N(s, t)$, for distinct s, t .

For $n \geq 1$, the n th term of R is the nonnegative chain integral

$$\int_{\mathbb{D}^{n+1}} P_s(y_0) \left(\prod_{j=0}^{n-1} G(y_j, y_{j+1}) \right) P_t(y_n) dq(y_0) \cdots dq(y_n).$$

Reversing the variables and using symmetry of G proves symmetry in s, t ; the zeroth term is also symmetric. The Poisson kernel satisfies $\int_{S^1} P_s(y) ds = 1$. Tonelli's theorem, applied to the nonnegative chains and series, therefore gives

$$\iint R(s, t) ds dt = \sum_{n \geq 0} \langle 1, T^n 1 \rangle_q \leq \sum_{n \geq 0} d^n \|1\|_q^2 = \frac{q(\mathbb{D})}{1-d}.$$

This proves equation (4.12). Values on the diagonal $s = t$ do not affect these integrals, since it has zero product arclength measure.

We next prove equation (4.13). The Neumann series in equation (4.8) gives

$$W_h(p) = Ah(p) + \sum_{n \geq 1} T^n Ah(p).$$

Indeed, the series for $(I - T)^{-1} Ah$ converges in $L^2(q)$, and evaluation after applying T is bounded by $|Tf(p)| \leq \|G(p, \cdot)\|_q \|f\|_q$. For nonnegative scalar h , inserting equation (4.6) into the iterated integrals and applying Tonelli gives

$$T^n Ah(p) = \int_{S^1} (T^n P_s)(p) h(s) ds \quad (n \geq 0).$$

Summing gives equation (4.13). For signed bounded h , all exchanges are absolutely justified by

$$\int_{S^1} K(p, s) |h(s)| ds \leq \|h\|_\infty \int_{S^1} K(p, s) ds < \infty,$$

using boundedness of $K(p, \cdot)$.

Finally, the operator Neumann series gives

$$\langle Ah, (I - T)^{-1} Ak \rangle_q = \sum_{n \geq 0} \langle Ah, T^n Ak \rangle_q.$$

For nonnegative scalar h, k , their Poisson representations and Tonelli yield

$$\langle Ah, T^n Ak \rangle_q = \iint h(s) k(t) \left(\int_{\mathbb{D}} P_s T^n P_t dq \right) ds dt.$$

Summing proves equation (4.14). For signed bounded data the required absolute domination follows from equation (4.12):

$$\iint R(s, t) |h(s) k(t)| ds dt \leq \|h\|_\infty \|k\|_\infty \frac{q(\mathbb{D})}{1-d} < \infty.$$

The vector formulas follow by applying the scalar formulas to each component and summing. $\square$

4.3 The boundary energy

We now return to the conformal measure equation (4.1). A vector boundary function $h : S^1 \rightarrow \mathbb{R}^2$ is called tangent if $h(s)$ is tangent to $\partial\Omega$ at $\Phi(s)$. Write $h = (h_1, h_2)$ in fixed Cartesian coordinates, and define $W_h = (W_{h_1}, W_{h_2})$, where each component is the scalar solution in equation (4.8). Set

$$X_h(y) = W_h(\Phi^{-1}(y)), \quad y \in \Omega.$$

Composition with the smooth diffeomorphism gives $X_h \in H^1(\Omega; \mathbb{R}^2)$, and its trace satisfies $\text{Tr } X_h(\Phi(s)) = h(s)$. Thus tangent data give tangent trace on $\partial\Omega$.

As in the preceding section on tangent vector fields, write

$$Q(X) = \int_{\Omega} (|\text{div } X|^2 + |\text{curl } X|^2) dy, \quad \text{curl } X = \partial_1 X_2 - \partial_2 X_1,$$

and define

$$E(h) = Q(X_h) - \lambda_N \|X_h\|_{L^2(\Omega)}^2, \quad E(h, k) = \frac{1}{4} (E(h+k) - E(h-k)).$$

The vector variational principle Proposition 3.1 gives $E(h) \geq 0$ for tangent Lipschitz data, with equality exactly when $X_h = \nabla v$ for some v in the real Neumann eigenspace V for λ_N . Let ν be the outward unit normal to $\partial\Omega$, $\tau = (-\nu_2, \nu_1)$, and $\kappa = \det(\tau, \partial_\tau \tau)$, as in that section.

Lemma 4.3. *For tangent Lipschitz data $h, k : S^1 \rightarrow \mathbb{R}^2$,*

$$\begin{aligned} E(h, k) &= \frac{1}{2} \iint_{S^1 \times S^1} N_0(s, t) (h(s) - h(t)) \cdot (k(s) - k(t)) ds dt \\ &\quad - \iint_{S^1 \times S^1} R(s, t) h(s) \cdot k(t) ds dt + \int_{S^1} |\Phi'| (\kappa \circ \Phi) h \cdot k ds. \end{aligned} \tag{4.18}$$

Proof. Polarising the curvature identity lemma 3.2 for tangent vector fields gives

$$E(h, k) = \int_{\Omega} \nabla X_h : \nabla X_k dy - \lambda_N \int_{\Omega} X_h \cdot X_k dy + \int_{\partial\Omega} \kappa \text{Tr } X_h \cdot \text{Tr } X_k d\sigma,$$

where $\nabla X_h : \nabla X_k = \sum_{j=1}^2 \nabla(X_h)_j \cdot \nabla(X_k)_j$. For each Cartesian component, conformality and change of variables give

$$\int_{\Omega} \nabla(X_h)_j \cdot \nabla(X_k)_j dy = \int_{\mathbb{D}} \nabla W_{h_j} \cdot \nabla W_{k_j} dx.$$

Indeed, the inverse derivative contributes $|\Phi'|^{-2}$ to the gradient product, the rotation preserves dot products, and the area Jacobian is $|\Phi'|^2$. Also

$$\lambda_N \int_{\Omega} X_h \cdot X_k dy = \int_{\mathbb{D}} W_h \cdot W_k \lambda_N |\Phi'|^2 dx = \langle W_h, W_k \rangle_q.$$

For the boundary integral, the parametrisation $\theta \mapsto \Phi(e^{i\theta})$ has speed

$$|\Phi'(e^{i\theta}) i e^{i\theta}| = |\Phi'(e^{i\theta})|.$$

Thus $d\sigma = |\Phi'(s)| ds$, and the trace identities give

$$\int_{\partial\Omega} \kappa \operatorname{Tr} X_h \cdot \operatorname{Tr} X_k d\sigma = \int_{S^1} |\Phi'|(\kappa \circ \Phi) h \cdot k ds.$$

It remains to express the two integrals over $\mathbb{D}$ in terms of boundary data. The difference $c_h = W_h - Ah$ has zero trace. Testing the equation for each component of W_k by the corresponding component of c_h gives

$$\int_{\mathbb{D}} \nabla c_h : \nabla W_k dx = \langle c_h, W_k \rangle_q.$$

Consequently

$$\int_{\mathbb{D}} \nabla W_h : \nabla W_k dx - \langle W_h, W_k \rangle_q = \int_{\mathbb{D}} \nabla Ah : \nabla W_k dx - \langle Ah, W_k \rangle_q.$$

Harmonicity of Ah and zero trace of $W_k - Ak$ give $\int_{\mathbb{D}} \nabla Ah : \nabla (W_k - Ak) dx = 0$. Since $W_k = (I - T)^{-1} Ak$ in $L^2(q)$, the last expression is

$$\int_{\mathbb{D}} \nabla Ah : \nabla Ak dx - \langle Ah, (I - T)^{-1} Ak \rangle_q.$$

The second term is the integral with R in equation (4.14).

For the harmonic term we prove

$$\int_{\mathbb{D}} \nabla Ah : \nabla Ak dx = \frac{1}{2\pi} \iint_{S^1 \times S^1} \frac{(h(s) - h(t)) \cdot (k(s) - k(t))}{|s - t|^2} ds dt. \quad (4.19)$$

First consider scalar Fourier modes $e_m(s) = s^m$, $m \in \mathbb{Z}$. For this calculation only, use complex functions and conjugate the second factor in each energy pairing. The harmonic extension is $Ae_m(re^{i\theta}) = r^{|m|} e^{im\theta}$, and direct integration gives

$$\int_{\mathbb{D}} |\nabla Ae_m|^2 dx = 2\pi|m|.$$

For $m \geq 1$,

$$\frac{s^m - t^m}{s - t} = \sum_{j=0}^{m-1} s^{m-1-j} t^j.$$

The summands are orthogonal in $L^2(S^1 \times S^1)$ and each has squared norm $(2\pi)^2$. Hence

$$\iint \frac{|s^m - t^m|^2}{|s - t|^2} ds dt = (2\pi)^2 m.$$

Complex conjugation gives the same formula with $|m|$ for negative m , and both energies vanish for $m = 0$. After multiplication by $1/(2\pi)$, this agrees with the disk energy.

Different modes have zero cross terms in the disk integral by angular orthogonality. For the boundary integral, set

$$J_{m\ell} = \iint \frac{(s^m - t^m)(\overline{s^\ell - t^\ell})}{|s - t|^2} ds dt.$$

Rotating both variables by $e^{i\alpha}$ preserves arclength and the denominator, so $J_{m\ell} = e^{i(m-\ell)\alpha} J_{m\ell}$ for every α . Thus $J_{m\ell} = 0$ when $m \neq \ell$. The matching mode energies and vanishing cross terms

prove equation (4.19) for real scalar trigonometric polynomials, and then for smooth data by convergence of their Fourier sums in $C^1(S^1)$.

For Lipschitz data, smooth periodic convolution gives approximants converging in $H^1(S^1)$ with uniformly bounded Lipschitz constants. The Poisson extensions converge in $H^1(\mathbb{D})$ by the Fourier energy formulas above. On the boundary, the integrands are dominated by a constant, since

$$\frac{|h(s) - h(t)| |k(s) - k(t)|}{|s - t|^2} \leq \text{Lip}(h) \text{Lip}(k).$$

Dominated convergence therefore proves the scalar identity for Lipschitz data. Summing over the two components proves equation (4.19) for vectors. Since $N_0(s, t) = 1/(\pi|s - t|^2)$, combining this identity with the preceding calculations gives equation (4.18). $\square$

Fix a nonzero $u \in V$, and define

$$g(s) = (\nabla u)(\Phi(s)), \quad s \in S^1. \quad (4.20)$$

Elliptic regularity [8, Chapter III, Theorems 6.4 and 6.5] gives $u \in C^\infty(\overline{\Omega})$, so g is smooth. It is tangent by the Neumann condition $\partial_\nu u = 0$. For each Cartesian derivative $v_j = \partial_j u$, differentiation of $-\Delta u = \lambda_N u$ gives $-\Delta v_j = \lambda_N v_j$. Conformal change of variables then gives

$$-\Delta(v_j \circ \Phi) = \lambda_N |\Phi'|^2 (v_j \circ \Phi) \quad \text{in } \mathbb{D}, \quad \text{Tr}(v_j \circ \Phi) = g_j.$$

Uniqueness in equation (4.7) therefore implies

$$W_g = (\nabla u) \circ \Phi, \quad X_g = \nabla u, \quad E(g) = 0. \quad (4.21)$$

The last equality is the equality case of the vector variational principle.

Proposition 4.4 (Multiplier identity). *For every real Lipschitz function b on S^1 ,*

$$E(bg) = \frac{1}{2} \iint_{S^1 \times S^1} N(s, t) (b(s) - b(t))^2 g(s) \cdot g(t) \, ds \, dt. \quad (4.22)$$

Proof. The products bg and b^2g are tangent Lipschitz boundary data. Since E is positive semidefinite on such data (by Proposition 3.1) and $E(g) = 0$, for every tangent Lipschitz h and every real t we have

$$0 \leq E(g + th) = 2tE(g, h) + t^2E(h).$$

If $E(g, h) \neq 0$, choosing t of the opposite sign and sufficiently small contradicts this inequality. Thus $E(g, h) = 0$ for all such h , and in particular $E(g, b^2g) = 0$. Consequently

$$E(bg) = E(bg, bg) - E(g, b^2g).$$

Apply equation (4.18) to both pairings on the right. The curvature integrals cancel because their pointwise factors are identical:

$$|bg|^2 = g \cdot (b^2g) = b^2|g|^2.$$

Write $b_s = b(s)$ and $g_s = g(s)$. For the harmonic integrals, expansion gives

$$\begin{aligned} & |b_s g_s - b_t g_t|^2 - (g_s - g_t) \cdot (b_s^2 g_s - b_t^2 g_t) \\ &= (b_s^2 + b_t^2 - 2b_s b_t) g_s \cdot g_t = (b_s - b_t)^2 g_s \cdot g_t. \end{aligned}$$

Their difference is therefore

$$\frac{1}{2} \iint N_0(s, t)(b_s - b_t)^2 g_s \cdot g_t \, ds \, dt.$$

For the integrals with R , subtraction gives

$$\iint R(s, t)(b_t^2 - b_s b_t) g_s \cdot g_t \, ds \, dt.$$

These integrals are absolutely convergent because b, g are bounded and R is integrable. Symmetry of R and $g_s \cdot g_t$ permits averaging the integrand with the one obtained by exchanging s, t . Since

$$\frac{1}{2}(b_t^2 - b_s b_t + b_s^2 - b_s b_t) = \frac{1}{2}(b_s - b_t)^2,$$

this contribution equals

$$\frac{1}{2} \iint R(s, t)(b_s - b_t)^2 g_s \cdot g_t \, ds \, dt.$$

Adding the two contributions and using $N = N_0 + R$ proves equation (4.22).

For completeness, the integral in that identity is absolutely convergent. If $L = \text{Lip}(b)$, then $|b_s - b_t| \leq L|s - t|$, so

$$|N_0(s, t)(b_s - b_t)^2 g_s \cdot g_t| \leq \frac{L^2}{\pi} \|g\|_\infty^2.$$

This bounds the integrand off the diagonal by a constant. For the remaining part,

$$|R(s, t)(b_s - b_t)^2 g_s \cdot g_t| \leq 4\|b\|_\infty^2 \|g\|_\infty^2 R(s, t),$$

which is integrable by equation (4.12). □

5 Reciprocal Green functions and a squared-distance kernel

Write $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and $S^1 = \partial\mathbb{D}$, and identify $\mathbb{C}$ with $\mathbb{R}^2$. We use the solution kernel K and the boundary kernel N constructed in section 4, under the measure assumptions restated below.

A real symmetric kernel H with zero diagonal is *conditionally negative semidefinite* if

$$\sum_{i,j=1}^n b_i b_j H(s_i, s_j) \leq 0 \quad \text{whenever} \quad \sum_{i=1}^n b_i = 0,$$

for every finite family of points s_i and real coefficients b_i .

Theorem 5.1. *Let $dq = a \, dx$ on $\mathbb{D}$, where $a \in L^\infty(\mathbb{D})$ is nonnegative, and suppose that there is a constant $0 < d < 1$ such that*

$$\int_{\mathbb{D}} |\psi|^2 \, dq \leq d \int_{\mathbb{D}} |\nabla \psi|^2 \, dx \quad (\psi \in H_0^1(\mathbb{D})).$$

For the associated kernels K, N and every $p \in \mathbb{D}$, the kernel

$$D_p(s, t) = \frac{K(p, s)K(p, t)}{N(s, t)} \quad (s \neq t), \quad D_p(s, s) = 0, \tag{5.1}$$

is conditionally negative semidefinite on S^1 .

Remark 5.2. When $q = 0$, one has $K(p, s) = P_s(p)$ and $N(s, t) = 1/(\pi|s - t|^2)$, where

$$P_s(p) = \frac{1 - |p|^2}{2\pi|s - p|^2}.$$

The disk automorphism $\phi_p(z) = (z - p)/(1 - \bar{p}z)$ maps p to zero and maps S^1 onto itself. Direct subtraction gives

$$|\phi_p(s) - \phi_p(t)|^2 = \frac{(1 - |p|^2)^2 |s - t|^2}{|s - p|^2 |t - p|^2} \quad (s, t \in S^1).$$

Consequently

$$D_p(s, t) = \pi P_s(p) P_t(p) |s - t|^2 = \frac{1}{4\pi} |\phi_p(s) - \phi_p(t)|^2.$$

Thus D_p is the squared Euclidean distance between the transformed boundary points, multiplied by $1/(4\pi)$.

For a real symmetric matrix M , let $n_+(M)$ denote its *positive index of inertia*: the number of strictly positive eigenvalues, counted with multiplicity. For a real symmetric kernel H on a set X , the assertion $n_+(H) \leq 1$ means that

$$n_+([H(x_i, x_j)]_{i,j=1}^n) \leq 1$$

for every finite set of distinct points $x_1, \dots, x_n \in X$. It is a condition on finite sampled matrices. Likewise, a complex kernel is positive semidefinite if every finite sampled matrix is Hermitian positive semidefinite. Reciprocals and powers of kernels below are entrywise, rather than operator inverses or compositions.

Definition 5.1 (Reciprocal matrix condition). Let H be a real symmetric kernel on X , with finite strictly positive values off the diagonal and strictly positive, possibly infinite, values on the diagonal. We say that H satisfies the *reciprocal matrix condition* if, for every finite set of distinct points $x_1, \dots, x_n \in X$,

$$n_+(Q_H) \leq 1, \quad (Q_H)_{ij} = \frac{1}{H(x_i, x_j)},$$

where $1/(+\infty) = 0$.

The condition in Definition 5.1 is related to the McCullough–Quiggin characterization of complete Nevanlinna–Pick kernels [27, 32]. For an irreducible positive semidefinite kernel with finite, nonzero entries, that characterization requires each finite reciprocal matrix to have exactly one positive eigenvalue [10, Corollary 1.12]. Here we use the weaker bound $n_+ \leq 1$, which also accommodates zero diagonal entries and limits. For a reciprocal-matrix condition in an earlier Green-function setting, see [32, Section 2.2, Corollary 2.2.2 and Theorem 2.2.3]. Invertible diagonal congruence preserves n_+ by Sylvester’s law of inertia. Subtracting a positive semidefinite matrix cannot increase n_+ : every subspace on which the difference is positive definite is also a positive subspace for the original matrix. The bound $n_+ \leq 1$ also passes to principal submatrices and entrywise limits of matrices of a fixed finite size. For the latter assertion, use continuity of the ordered eigenvalues of a real symmetric matrix with respect to its entries.

Lemma 5.A. *Let B, C be real symmetric positive semidefinite $n \times n$ matrices, with $\text{rank } C \leq 1$. Then*

$$n_+(C - B) \leq 1.$$

In particular, $n_+(uu^T - B) \leq 1$ for every $u \in \mathbb{R}^n$.

Proof. For $b \in \ker C$,

$$b^T(C - B)b = -b^T B b \leq 0.$$

Since $\ker C$ has codimension at most one, every subspace of dimension at least two meets it nontrivially. Thus $C - B$ cannot be positive definite on a two-dimensional subspace, so it has at most one positive eigenvalue. The particular case follows because uu^T is positive semidefinite and has rank at most one. $\square$

5.1 A bounded regularization of the Green kernel

The Dirichlet Green function of the disk is

$$G(x, y) = \frac{1}{4\pi} \log \left(1 + \frac{(1 - |x|^2)(1 - |y|^2)}{|x - y|^2} \right) \quad (x \neq y),$$

with $G(x, x) = +\infty$. A quadrature formula sampling G at its integration nodes would encounter these infinite diagonal values. We therefore construct bounded replacements with finite positive diagonal satisfying the condition in Definition 5.1.

Lemma 5.3. *There are bounded continuous strictly positive kernels C_ϵ on $\mathbb{D} \times \mathbb{D}$, $0 < \epsilon \leq 1$, such that $C_\epsilon \uparrow G$ as $\epsilon \downarrow 0$, including on the diagonal, and each C_ϵ satisfies the reciprocal matrix condition in Definition 5.1.*

Proof. Define the two-variable kernel

$$h(x, y) = \frac{|x - y|^2}{(1 - |x|^2)(1 - |y|^2)}.$$

For a scalar argument $\tau \geq 0$, define

$$L(\tau) = \frac{1}{\log(1 + 1/\tau)} \quad (\tau > 0), \quad L(0) = 0, \quad f(\tau) = \tau + \frac{1}{2} - L(\tau).$$

Both scalar functions are continuous on $[0, \infty)$, and

$$\frac{1}{G(x, y)} = 4\pi L(h(x, y)),$$

with reciprocal zero on the diagonal. We will show that $n_+(h + \frac{1}{2}) \leq 1$ and that the kernel $(x, y) \mapsto f(h(x, y))$ is positive semidefinite.

For a finite set $x_1, \dots, x_n$, put $A_{ij} = h(x_i, x_j) + \frac{1}{2}$ and $W = \text{diag}(1 - |x_i|^2)$. Expanding $|x_i - x_j|^2$ gives

$$(WAW)_{ij} = \frac{1}{2}(1 + |x_i|^2)(1 + |x_j|^2) - 2x_i \cdot x_j. \quad (5.2)$$

The first term is the rank-one matrix uu^T , where $u_i = (1 + |x_i|^2)/\sqrt{2}$, and the matrix $[2x_i \cdot x_j]$ is a positive semidefinite Gram matrix. By Lemma 5.A, $n_+(WAW) \leq 1$. Since W is invertible, Sylvester's law of inertia gives $n_+(A) \leq 1$.

For $\tau > 0$, direct integration gives

$$\begin{aligned} \int_0^1 \tau^{1-\alpha} (\tau + 1)^\alpha d\alpha &= \tau \int_0^1 \exp(\alpha \log(1 + 1/\tau)) d\alpha \\ &= \frac{\tau((1 + 1/\tau) - 1)}{\log(1 + 1/\tau)} = L(\tau). \end{aligned} \quad (5.3)$$

For related integral representations of the weighted geometric mean and the logarithmic mean, see [12]. We derive the further representation needed here.

For $0 < \alpha < 1$, let V_α be a random variable with probability density

$$c_\alpha \frac{v^{\alpha-1}}{1+v} \quad (v > 0), \quad c_\alpha = \left(\int_0^\infty \frac{v^{\alpha-1}}{1+v} dv \right)^{-1}.$$

The normalizing integral is finite at both zero and infinity. Set $R_\alpha = (1 + V_\alpha)^{-1} \in (0, 1)$, and write $\mathbb{E}_\alpha$ for expectation under this law. The sans-serif symbols denote random variables; v and r denote fixed scalar values. Integration by parts, with vanishing endpoint terms, gives

$$\mathbb{E}_\alpha(1 - R_\alpha) = c_\alpha \int_0^\infty \frac{v^\alpha}{(1+v)^2} dv = \alpha c_\alpha \int_0^\infty \frac{v^{\alpha-1}}{1+v} dv = \alpha.$$

For $\tau > 0$, substituting $w = v\tau/(\tau + 1)$ in the integral below gives

$$\begin{aligned} \tau^{1-\alpha}(\tau + 1)^\alpha &= c_\alpha \int_0^\infty \frac{\tau(\tau + 1)}{(\tau + 1) + v\tau} v^{\alpha-1} dv \\ &= \mathbb{E}_\alpha \frac{\tau(\tau + 1)}{\tau + R_\alpha} \\ &= \mathbb{E}_\alpha \left[\tau + (1 - R_\alpha) - \frac{R_\alpha(1 - R_\alpha)}{\tau + R_\alpha} \right]. \end{aligned} \tag{5.4}$$

At $\tau = 0$ the last two expressions also agree, and both are zero. Integrating in α and using equation (5.3) therefore yields

$$f(\tau) = \int_0^1 \mathbb{E}_\alpha \frac{R_\alpha(1 - R_\alpha)}{\tau + R_\alpha} d\alpha, \quad 0 \leq f(\tau) \leq f(0) = \frac{1}{2}. \tag{5.5}$$

The formula at zero follows directly from $\mathbb{E}_\alpha(1 - R_\alpha) = \alpha$ and $f(0) = 1/2$. The bound follows from $R_\alpha/(\tau + R_\alpha) \leq 1$.

We next prove the positive semidefiniteness needed in this mixture. We use the standard Gram-kernel criterion and the Schur product theorem [11, Propositions 17.1 and 17.5]. In particular, a pointwise convergent sum $\sum_m \varphi_m(x) \overline{\varphi_m(y)}$ is positive semidefinite: its finite partial sums are Gram kernels, and the cone of positive semidefinite matrices is closed. More generally, finite-valued pointwise limits of positive semidefinite kernels are positive semidefinite; see also [11, Problem 17.4].

The power series

$$S(x, y) = \frac{1}{1 - x\bar{y}} = \sum_{m=0}^\infty x^m \bar{y}^m \quad (x, y \in \mathbb{D})$$

satisfies this criterion. Its complex conjugate is also positive semidefinite, by the same criterion with $\varphi_m(x) = \bar{x}^m$. The Schur product theorem shows that $S(x, y) \overline{S(x, y)} = |1 - x\bar{y}|^{-2}$ is positive semidefinite. On every finite set, multiplying this kernel by $(1 - |x|^2)(1 - |y|^2)$ is congruence by the positive diagonal matrix $\text{diag}(1 - |x_i|^2)$ and preserves positive semidefiniteness. Since

$$|1 - x\bar{y}|^2 = |x - y|^2 + (1 - |x|^2)(1 - |y|^2),$$

the resulting kernel is

$$k(x, y) = \frac{1}{1 + h(x, y)} = \frac{(1 - |x|^2)(1 - |y|^2)}{|1 - x\bar{y}|^2}.$$

Thus k is positive semidefinite and $0 < k(x, y) \leq 1$. For each fixed scalar $0 < r < 1$, the geometric series gives

$$\frac{1}{h(x, y) + r} = \frac{k(x, y)}{1 - (1 - r)k(x, y)} = \sum_{m=0}^{\infty} (1 - r)^m k(x, y)^{m+1}. \quad (5.6)$$

Every positive integer entrywise power of k is positive semidefinite by repeated application of the Schur product theorem. The coefficients in this series are nonnegative, so its partial sums are positive semidefinite. The series converges everywhere, including when $x = y$, because $(1 - r)k(x, y) \leq 1 - r < 1$. For each fixed finite set of points, the matrices of partial sums converge entrywise. Their limit is positive semidefinite: for every vector b , its quadratic form is the limit of the nonnegative quadratic forms of the partial sums. Thus $(x, y) \mapsto 1/(h(x, y) + r)$ is positive semidefinite.

For any finite points x_i and real coefficients b_i , the mixture equation (5.5) now gives

$$\begin{aligned} \sum_{i,j} b_i b_j f(h(x_i, x_j)) &= \int_0^1 \mathbb{E}_\alpha \left[R_\alpha (1 - R_\alpha) \sum_{i,j} \frac{b_i b_j}{h(x_i, x_j) + R_\alpha} \right] d\alpha \\ &\geq 0. \end{aligned}$$

For each realization of $R_\alpha \in (0, 1)$, the inner quadratic form is nonnegative by the preceding argument. The finite sum may be interchanged with the integrals because

$$0 \leq \frac{R_\alpha (1 - R_\alpha)}{h(x_i, x_j) + R_\alpha} \leq 1 - R_\alpha \leq 1.$$

These integrands are measurable under the explicit densities above and integrable in both v and α . This proves that $(x, y) \mapsto f(h(x, y))$ is positive semidefinite, including on the diagonal.

Define C_ϵ by

$$\begin{aligned} \frac{1}{C_\epsilon(x, y)} &= 4\pi \left[h(x, y) + \frac{1}{2} - (1 - \epsilon)f(h(x, y)) \right] \\ &= 4\pi [L(h(x, y)) + \epsilon f(h(x, y))]. \end{aligned} \quad (5.7)$$

On any finite set, its reciprocal matrix is 4π times the matrix of $h + \frac{1}{2}$, minus $4\pi(1 - \epsilon)$ times the matrix of $f \circ h$. For the former matrix A , equation (5.2) shows that WAW is a rank-one positive semidefinite matrix minus a positive semidefinite Gram matrix. Hence $n_+(WAW) \leq 1$ by Lemma 5.A. Since W is invertible, Sylvester's law of inertia gives $n_+(A) = n_+(WAW) \leq 1$. The matrix of $f \circ h$ is positive semidefinite by the mixture argument. Since subtracting a positive semidefinite matrix cannot increase n_+ , the reciprocal matrix has at most one positive eigenvalue.

For $\tau \geq 0$,

$$L(\tau) + \epsilon f(\tau) = \epsilon \left(\tau + \frac{1}{2} \right) + (1 - \epsilon)L(\tau) \geq \frac{\epsilon}{2}.$$

The reciprocal in equation (5.7) is therefore at least $2\pi\epsilon$. Continuity of h, L, f shows that C_ϵ is continuous and strictly positive, with $C_\epsilon \leq 1/(2\pi\epsilon)$. Finally $f \geq 0$ implies $C_\epsilon \uparrow G$ as $\epsilon \downarrow 0$ off the diagonal, and $0 < C_\epsilon \leq G$. On the diagonal, $C_\epsilon(x, x) = 1/(2\pi\epsilon) \uparrow +\infty = G(x, x)$. $\square$

5.2 A finite resolvent update

Lemma 5.4. *Let M be a real symmetric $n \times n$ matrix with strictly positive entries, and suppose that its entrywise reciprocal $Q = [1/M_{ij}]_{i,j=1}^n$ satisfies $n_+(Q) \leq 1$. For any index $z \in \{1, \dots, n\}$*

and any scalar $c \geq 0$, define

$$\widetilde{M}_{ij} = M_{ij} + cM_{iz}M_{zj}.$$

Then the entrywise reciprocal $\widetilde{Q} = [1/\widetilde{M}_{ij}]_{i,j=1}^n$ also satisfies $n_+(\widetilde{Q}) \leq 1$.

Proof. The case $c = 0$ is immediate, since $\widetilde{M} = M$. The assertion is also immediate when $n = 1$. Hence assume $c > 0$ and $n \geq 2$. Define the diagonal matrix

$$D = \text{diag}(M_{1z}, \dots, M_{nz})$$

and set $H = DQD$. All diagonal entries of D are positive, so D is invertible and Sylvester's law of inertia gives $n_+(H) = n_+(Q) \leq 1$. By symmetry of M ,

$$H_{ij} = \frac{M_{iz}M_{jz}}{M_{ij}} = \frac{M_{iz}M_{zj}}{M_{ij}}.$$

Write $\gamma = M_{zz} > 0$. The z -row and z -column of H are constant, because $H_{iz} = M_{iz}M_{zz}/M_{iz} = \gamma$. After simultaneously reordering the rows and columns to place index z last, let

$$H_0 = [H_{ij}]_{i,j \neq z}$$

be the $(n-1) \times (n-1)$ principal submatrix obtained by deleting the z -row and z -column. With $\mathbf{1}_{n-1}$ denoting the all-ones vector of length $n-1$, the full matrix is

$$H = \begin{pmatrix} H_0 & \gamma \mathbf{1}_{n-1} \\ \gamma \mathbf{1}_{n-1}^T & \gamma \end{pmatrix}.$$

Changing from the standard basis to $e_1 - e_n, \dots, e_{n-1} - e_n, e_n$ gives an invertible congruence transforming H into

$$\begin{pmatrix} H_0 - \gamma \mathbf{1}_{n-1} \mathbf{1}_{n-1}^T & 0 \\ 0 & \gamma \end{pmatrix}.$$

The last block already contributes one positive eigenvalue. Since $n_+(H) \leq 1$, the first block must be negative semidefinite. Consequently the full $n \times n$ matrix

$$U = \gamma \mathbf{1}_n \mathbf{1}_n^T - H = \begin{pmatrix} \gamma \mathbf{1}_{n-1} \mathbf{1}_{n-1}^T - H_0 & 0 \\ 0 & 0 \end{pmatrix}$$

is positive semidefinite. Here $\mathbf{1}_n$ is the all-ones vector of length n . Since $H_{ii} > 0$,

$$0 \leq U_{ii} = \gamma - H_{ii} < \gamma.$$

The Cauchy-Schwarz inequality for a positive semidefinite matrix then gives

$$|U_{ij}|^2 \leq U_{ii}U_{jj} < \gamma^2, \quad |U_{ij}| < \gamma.$$

For the reciprocal matrix of the updated matrix $\widetilde{M}$, use the same diagonal matrix D and define $\widetilde{H} = D\widetilde{Q}D$. Its entries are

$$\widetilde{H}_{ij} = \frac{M_{iz}M_{zj}}{M_{ij} + cM_{iz}M_{zj}} = \frac{H_{ij}}{1 + cH_{ij}}.$$

Define the constant rank-one positive semidefinite matrix

$$C_c = \frac{\gamma}{1 + c\gamma} \mathbf{1}_n \mathbf{1}_n^T.$$

Using $H_{ij} = \gamma - U_{ij}$, the entries of $B_c = C_c - \tilde{H}$ satisfy

$$\begin{aligned} (B_c)_{ij} &= \frac{\gamma}{1 + c\gamma} - \frac{H_{ij}}{1 + cH_{ij}} \\ &= \frac{U_{ij}}{(1 + c\gamma)(1 + c\gamma - cU_{ij})} \\ &= \sum_{m=0}^{\infty} \frac{c^m U_{ij}^{m+1}}{(1 + c\gamma)^{m+2}}. \end{aligned} \tag{5.8}$$

The series converges absolutely in every entry, because $|cU_{ij}| < c\gamma < 1 + c\gamma$. Each positive integer entrywise power of U is positive semidefinite by the Schur product theorem [11, Proposition 17.5]; this conclusion does not require the entries of U to be nonnegative. All coefficients in equation (5.8) are nonnegative, so the matrices of partial sums are positive semidefinite. Their entrywise limit B_c is positive semidefinite, since each quadratic form of B_c is the limit of the corresponding nonnegative quadratic forms.

Thus $\tilde{H} = C_c - B_c$ is a rank-one positive semidefinite matrix minus a positive semidefinite matrix. By Lemma 5.A, $n_+(\tilde{H}) \leq 1$. Since D is invertible, Sylvester's law of inertia gives $n_+(\tilde{Q}) = n_+(\tilde{H}) \leq 1$, as required. $\square$

5.3 The interior resolvent kernel

Let $0 \leq \rho \leq q$ be supported on a compact subset of $\mathbb{D}$. Write $\langle f, g \rangle_\rho = \int f g d\rho$ and $\|f\|_\rho = \langle f, f \rangle_\rho^{1/2}$ on the real space $L^2(\rho)$. The Green operator

$$T_\rho f(x) = \int_{\mathbb{D}} G(x, y) f(y) d\rho(y)$$

has $\|T_\rho\| \leq d < 1$, by the Green-operator estimate of section 4. For each interior point x , the section $G(x, \cdot)$ belongs to $L^2(\rho)$: the density of ρ is bounded, and the logarithmic singularity is locally square integrable. Define

$$\begin{aligned} M_\rho(x, y) &= G(x, y) + \langle G(x, \cdot), (I - T_\rho)^{-1} G(y, \cdot) \rangle_\rho \quad (x \neq y), \\ M_\rho(x, x) &= +\infty. \end{aligned} \tag{5.9}$$

Proposition 5.5. *The kernel M_ρ satisfies the reciprocal matrix condition in Definition 5.1. In particular, every finite reciprocal matrix has zero diagonal and at most one positive eigenvalue.*

Proof. First fix $0 < \epsilon \leq 1$ and define the operator

$$T_{\rho, \epsilon} f(x) = \int_{\mathbb{D}} C_\epsilon(x, y) f(y) d\rho(y)$$

on $L^2(\rho)$. The bounds $0 < C_\epsilon \leq G$ from lemma 5.3 give the pointwise inequality

$$|T_{\rho, \epsilon} f(x)| \leq T_\rho |f|(x) \quad \text{for } \rho\text{-almost every } x.$$

Taking $L^2(\rho)$ norms gives

$$\|T_{\rho,\epsilon}f\|_\rho \leq \|T_\rho|f|\|_\rho \leq d\|f\|_\rho, \quad \|T_{\rho,\epsilon}\| \leq d < 1.$$

Hence $I - T_{\rho,\epsilon}$ is invertible. For every $x, y \in \mathbb{D}$, including $x = y$, define

$$\begin{aligned} M_{\rho,\epsilon}(x, y) &= C_\epsilon(x, y) \\ &\quad + \langle C_\epsilon(x, \cdot), (I - T_{\rho,\epsilon})^{-1}C_\epsilon(y, \cdot) \rangle_\rho. \end{aligned} \quad (5.10)$$

The sections in this formula belong to $L^2(\rho)$ because C_ϵ is bounded and ρ is finite. Thus both the kernel sections and the operator in equation (5.9) have been replaced by their C_ϵ counterparts.

We first prove the condition in Definition 5.1 for $M_{\rho,\epsilon}$ by quadrature. If $\rho = 0$, then $M_{\rho,\epsilon} = C_\epsilon$, and this follows directly from lemma 5.3. For the quadrature argument assume $\rho \neq 0$, and fix a nonempty compact set $E \subset \mathbb{D}$ such that $\rho(\mathbb{D} \setminus E) = 0$. For each integer $m \geq 1$, choose a finite Borel partition

$$E = \bigsqcup_{j=1}^{J_m} E_{m,j}, \quad \max_j \text{diam}(E_{m,j}) \leq \frac{1}{m},$$

with nonempty cells. Such partitions are obtained by intersecting E with a grid of half-open squares of side $1/(\sqrt{2}m)$ and discarding empty intersections. For each $j \in \{1, \dots, J_m\}$ choose $z_{m,j} \in E_{m,j}$, and put

$$w_{m,j} = \rho(E_{m,j}), \quad I_m = \{j : w_{m,j} > 0\}, \quad \zeta_m(x) = z_{m,j} \quad (x \in E_{m,j}).$$

The representatives are distinct because the cells are disjoint. Only indices in I_m contribute to quadrature sums, but choosing representatives also in the zero-mass cells defines ζ_m on all of E . In particular, $\sup_{x \in E} |\zeta_m(x) - x| \leq 1/m$.

Define the quadrature kernel and its operator by

$$\begin{aligned} C_{\epsilon,m}(x, y) &= C_\epsilon(\zeta_m(x), \zeta_m(y)) \quad (x, y \in E), \\ T_{\rho,\epsilon,m}f(x) &= \int_E C_{\epsilon,m}(x, y)f(y) d\rho(y) \quad (x \in E). \end{aligned}$$

Uniform continuity of C_ϵ on $E \times E$ gives

$$\delta_m = \sup_{x,y \in E} |C_{\epsilon,m}(x, y) - C_\epsilon(x, y)| \longrightarrow 0. \quad (5.11)$$

For $f \in L^2(\rho)$, Cauchy–Schwarz gives

$$|(T_{\rho,\epsilon,m} - T_{\rho,\epsilon})f(x)| \leq \delta_m \int_E |f| d\rho \leq \delta_m \sqrt{\rho(E)} \|f\|_\rho.$$

Taking the $L^2(\rho)$ norm in x yields the exact estimate

$$\|T_{\rho,\epsilon,m} - T_{\rho,\epsilon}\| \leq \rho(E)\delta_m \longrightarrow 0. \quad (5.12)$$

All operator norms in this argument are on $L^2(\rho)$ unless another space is specified. Choose m_0 so that $\rho(E)\delta_m \leq (1-d)/2$ for $m \geq m_0$. Then

$$\|T_{\rho,\epsilon,m}\| \leq \frac{1+d}{2} < 1, \quad \|(I - T_{\rho,\epsilon,m})^{-1}\| \leq \frac{2}{1-d}. \quad (5.13)$$

The resolvent identity

$$\begin{aligned} & (I - T_{\rho,\epsilon,m})^{-1} - (I - T_{\rho,\epsilon})^{-1} \\ &= (I - T_{\rho,\epsilon,m})^{-1} (T_{\rho,\epsilon,m} - T_{\rho,\epsilon}) (I - T_{\rho,\epsilon})^{-1} \end{aligned}$$

therefore gives

$$\|(I - T_{\rho,\epsilon,m})^{-1} - (I - T_{\rho,\epsilon})^{-1}\| \leq \frac{2\rho(E)\delta_m}{(1-d)^2} \longrightarrow 0. \quad (5.14)$$

For each fixed evaluation point $x \in \mathbb{D}$, define

$$u_{m,x}(z) = C_\epsilon(x, \zeta_m(z)), \quad u_x(z) = C_\epsilon(x, z) \quad (z \in E).$$

Uniform continuity on $\{x\} \times E$ gives

$$\|u_{m,x} - u_x\|_\rho \leq \sqrt{\rho(E)} \sup_{z \in E} |C_\epsilon(x, \zeta_m(z)) - C_\epsilon(x, z)| \longrightarrow 0. \quad (5.15)$$

For $m \geq m_0$, define

$$\begin{aligned} M_{\rho,\epsilon,m}(x, y) &= C_\epsilon(x, y) \\ &\quad + \langle u_{m,x}, (I - T_{\rho,\epsilon,m})^{-1} u_{m,y} \rangle_\rho. \end{aligned} \quad (5.16)$$

The bounds and convergences in equations (5.13) to (5.15) imply, for every fixed $x, y \in \mathbb{D}$,

$$M_{\rho,\epsilon,m}(x, y) \longrightarrow M_{\rho,\epsilon}(x, y). \quad (5.17)$$

Indeed, split the difference of the pairings by changing first $u_{m,x}$, then the inverse, then $u_{m,y}$. Cauchy–Schwarz bounds the resulting terms by the corresponding source and operator errors times bounded norms. Thus (5.17) is pointwise convergence in the evaluation points.

We now express equation (5.16) as a finite matrix formula. For this fixed m , let $N_m = |I_m|$, order the positive-mass cells by increasing index, and define

$$e_{m,j} = \frac{\mathbf{1}_{E_{m,j}}}{\sqrt{w_{m,j}}} \quad (j \in I_m), \quad \mathcal{S}_m = \text{span}\{\mathbf{1}_{E_{m,j}} : j \in I_m\} \subset L^2(\rho).$$

The subspace $\mathcal{S}_m$ consists of functions constant ρ -almost everywhere on each partition cell. The vectors $e_{m,j}$ form an orthonormal basis of this subspace, because the cells are disjoint and $\|\mathbf{1}_{E_{m,j}}\|_\rho^2 = w_{m,j}$. Define the isometry

$$J_m : \mathbb{R}^{N_m} \rightarrow L^2(\rho), \quad J_m b = \sum_{j \in I_m} b_j e_{m,j}.$$

Its range is $\mathcal{S}_m$, and $(J_m^* f)_j = w_{m,j}^{-1/2} \int_{E_{m,j}} f d\rho$. Define the symmetric matrix A_m and coordinate vectors $\xi_{m,x}$ by

$$(A_m)_{ij} = \sqrt{w_{m,i}} C_\epsilon(z_{m,i}, z_{m,j}) \sqrt{w_{m,j}}, \quad (\xi_{m,x})_j = \sqrt{w_{m,j}} C_\epsilon(x, z_{m,j}).$$

Direct substitution gives

$$T_{\rho,\epsilon,m} = J_m A_m J_m^*, \quad u_{m,x} = J_m \xi_{m,x}. \quad (5.18)$$

Thus the operator is represented by A_m on $\mathcal{S}_m$ and vanishes on $\mathcal{S}_m^\perp$. In particular,

$$\|A_m\|_{\mathbb{R}^{N_m} \rightarrow \mathbb{R}^{N_m}} = \|T_{\rho, \epsilon, m}\| \leq \frac{1+d}{2} < 1.$$

Since $(I - T_{\rho, \epsilon, m})J_m = J_m(I - A_m)$, equation (5.18) yields the exact formula

$$M_{\rho, \epsilon, m}(x, y) = C_\epsilon(x, y) + \xi_{m, x}^T (I - A_m)^{-1} \xi_{m, y}. \quad (5.19)$$

The Neumann series $(I - A_m)^{-1} = \sum_{k=0}^{\infty} A_m^k$ converges in matrix operator norm. Expanding the matrix products gives

$$\begin{aligned} & \xi_{m, x}^T (I - A_m)^{-1} \xi_{m, y} \\ &= \sum_{k=0}^{\infty} \sum_{j_0, \dots, j_k \in I_m} \left(\prod_{\ell=0}^k w_{m, j_\ell} \right) C_\epsilon(x, z_{m, j_0}) \\ & \quad \times \left(\prod_{\ell=0}^{k-1} C_\epsilon(z_{m, j_\ell}, z_{m, j_{\ell+1}}) \right) C_\epsilon(z_{m, j_k}, y). \end{aligned} \quad (5.20)$$

For $k=0$ the middle product is one. The intermediate indices may repeat, and each occurrence contributes its quadrature weight. All summands are nonnegative. Finiteness follows from (5.19), or directly from

$$|\xi_{m, x}^T A_m^k \xi_{m, y}| \leq \|\xi_{m, x}\| \|\xi_{m, y}\| \|A_m\|^k.$$

For a subset $J \subseteq I_m$, let $\mathcal{R}_{m, J}(x, y)$ denote the series on the right-hand side of (5.20) with every intermediate index $j_0, \dots, j_k$ restricted to J ; put $\mathcal{R}_{m, \emptyset} = 0$. Nonnegativity gives

$$0 \leq \mathcal{R}_{m, J}(x, y) \leq \mathcal{R}_{m, I_m}(x, y) = \xi_{m, x}^T (I - A_m)^{-1} \xi_{m, y} < \infty. \quad (5.21)$$

Define

$$M^{(J)}(x, y) = C_\epsilon(x, y) + \mathcal{R}_{m, J}(x, y).$$

In particular, $M^{(\emptyset)} = C_\epsilon$ and $M^{(I_m)} = M_{\rho, \epsilon, m}$.

Choose $j_* \in I_m \setminus J$, and write $z = z_{m, j_*}$ and $w = w_{m, j_*} > 0$. We calculate $M^{(J \cup \{j_*\})}$ by grouping the new terms according to the number $k \geq 1$ of occurrences of j_* among their intermediate indices. Splitting the sequence at these occurrences leaves $k+1$ segments whose intermediate indices belong to J . Summing over each segment gives a factor $M^{(J)}$; its direct C_ϵ term accounts for an empty intermediate sequence. Each of the k occurrences contributes a factor w . Therefore, by regrouping nonnegative series,

$$\begin{aligned} M^{(J \cup \{j_*\})}(x, y) &= M^{(J)}(x, y) \\ &+ \sum_{k=1}^{\infty} w^k M^{(J)}(x, z) (M^{(J)}(z, z))^{k-1} M^{(J)}(z, y). \end{aligned} \quad (5.22)$$

Every $M^{(J)}$ value is finite and strictly positive by (5.21). The series in (5.22) is finite by the same bound, so its geometric ratio satisfies $wM^{(J)}(z, z) < 1$. Summing it gives

$$\begin{aligned} M^{(J \cup \{j_*\})}(x, y) &= M^{(J)}(x, y) + c_J M^{(J)}(x, z) M^{(J)}(z, y), \\ c_J &= \frac{w}{1 - wM^{(J)}(z, z)} > 0. \end{aligned} \quad (5.23)$$

Fix a finite set of distinct evaluation points, and form its union with the positive-mass quadrature nodes, listed without repetitions. Sample every $M^{(J)}$ on this same finite set. Order the indices in I_m as $j_1, \dots, j_{N_m}$ and put $J_0 = \emptyset$, $J_\ell = \{j_1, \dots, j_\ell\}$. For J_0 , the sampled matrix is that of C_ϵ , whose reciprocal has at most one positive eigenvalue by lemma 5.3. At each step, (5.23) is exactly the update in lemma 5.4, with z an index in the sampled matrix. Applying that lemma successively for $\ell = 1, \dots, N_m$ proves the same bound for the reciprocal matrix of $M^{(I_m)} = M_{\rho, \epsilon, m}$. Taking the principal submatrix on the evaluation points proves that $M_{\rho, \epsilon, m}$ satisfies the condition in Definition 5.1.

By (5.17), for any fixed finite set $x_1, \dots, x_n$, $M_{\rho, \epsilon, m}(x_i, x_j) \rightarrow M_{\rho, \epsilon}(x_i, x_j)$. The approximating kernels are at least C_ϵ , by (5.20), so the limiting entries are strictly positive. Hence their reciprocal matrices converge entrywise as well. The bound $n_+ \leq 1$ passes to this limit of matrices of fixed size. Thus $M_{\rho, \epsilon}$ satisfies the condition in Definition 5.1 for each ϵ .

Finally let $\epsilon \downarrow 0$. For $k \geq 0$, define

$$\begin{aligned} \mathcal{I}_{\rho, \epsilon, k}(x, y) &= \int_{E^{k+1}} C_\epsilon(x, z_0) \\ &\quad \times \left(\prod_{\ell=0}^{k-1} C_\epsilon(z_\ell, z_{\ell+1}) \right) C_\epsilon(z_k, y) d\rho(z_0) \cdots d\rho(z_k). \end{aligned} \quad (5.24)$$

The middle product is one when $k = 0$. If $\rho = 0$, take any nonempty compact $E \subset \mathbb{D}$; all these integrals are zero. The operator Neumann series and the nonnegative integral kernels give

$$M_{\rho, \epsilon}(x, y) - C_\epsilon(x, y) = \sum_{k=0}^{\infty} \mathcal{I}_{\rho, \epsilon, k}(x, y). \quad (5.25)$$

By lemma 5.3, $C_\epsilon \uparrow G$. Thus each integrand in (5.24) increases pointwise to the product with G in place of every C_ϵ factor. Monotone convergence for the integrals and then their series gives

$$\begin{aligned} \lim_{\epsilon \downarrow 0} \sum_{k=0}^{\infty} \mathcal{I}_{\rho, \epsilon, k}(x, y) &= \sum_{k=0}^{\infty} \mathcal{I}_{\rho, G, k}(x, y) \\ &= \langle G(x, \cdot), (I - T_\rho)^{-1} G(y, \cdot) \rangle_\rho, \end{aligned} \quad (5.26)$$

where $\mathcal{I}_{\rho, G, k}$ is the same integral with G . The last equality follows from the Neumann series for $(I - T_\rho)^{-1}$; its k th pairing is precisely $\mathcal{I}_{\rho, G, k}$.

The correction in (5.26) is finite for every $x, y \in \mathbb{D}$, including $x = y$, because

$$\left| \langle G(x, \cdot), (I - T_\rho)^{-1} G(y, \cdot) \rangle_\rho \right| \leq \frac{\|G(x, \cdot)\|_\rho \|G(y, \cdot)\|_\rho}{1 - d}. \quad (5.27)$$

For $x \neq y$, the direct term also has a finite limit, so

$$M_{\rho, \epsilon}(x, y) \longrightarrow M_\rho(x, y) < \infty.$$

On the diagonal, the nonnegative correction gives

$$M_{\rho, \epsilon}(x, x) \geq C_\epsilon(x, x) = \frac{1}{2\pi\epsilon} \longrightarrow +\infty.$$

Thus, for any fixed finite set of distinct points $x_1, \dots, x_n$, the reciprocal matrices converge entrywise to the matrix with off-diagonal entries $1/M_\rho(x_i, x_j)$ and diagonal entries zero. Each approximating matrix has at most one positive eigenvalue, so the same is true of this limit. This is exactly the condition in Definition 5.1 for M_ρ . $\square$

5.4 The boundary limit

We first record explicit integral formulas for the boundary kernels. For any $0 \leq \rho \leq q$ and $k \geq 1$, define

$$\begin{aligned} \mathcal{K}_{\rho,k}(p, s) &= \int_{\mathbb{D}^k} G(p, x_1) \\ &\quad \times \left(\prod_{\ell=1}^{k-1} G(x_\ell, x_{\ell+1}) \right) P_s(x_k) d\rho(x_1) \cdots d\rho(x_k). \end{aligned} \quad (5.28)$$

For $k \geq 0$, define

$$\begin{aligned} \mathcal{R}_{\rho,k}(s, t) &= \int_{\mathbb{D}^{k+1}} P_s(x_0) \\ &\quad \times \left(\prod_{\ell=0}^{k-1} G(x_\ell, x_{\ell+1}) \right) P_t(x_k) d\rho(x_0) \cdots d\rho(x_k). \end{aligned} \quad (5.29)$$

Empty products in these formulas equal one. The nonnegative integrals and series define

$$\begin{aligned} K_\rho(p, s) &= P_s(p) + \sum_{k=1}^{\infty} \mathcal{K}_{\rho,k}(p, s), \\ R_\rho(s, t) &= \sum_{k=0}^{\infty} \mathcal{R}_{\rho,k}(s, t), \end{aligned} \quad (5.30)$$

$$N_\rho(s, t) = N_0(s, t) + R_\rho(s, t), \quad N_0(s, t) = \frac{1}{\pi|s - t|^2} \quad (s \neq t).$$

These are the definitions from section 4 with q replaced by ρ . In particular $K_q = K$, $R_q = R$, and $N_q = N$. The integrals are understood as nonnegative integrals before finiteness is asserted; they do not require a Poisson section to belong to $L^2(q)$.

Lemma 5.6. *Let $s_1, \dots, s_n \in S^1$ be distinct, and let $p \in \mathbb{D}$. Define the real symmetric matrix*

$$\begin{aligned} \mathcal{B} &= \begin{pmatrix} B & \ell \\ \ell^T & 0 \end{pmatrix}, \\ B_{ij} &= \begin{cases} 1/N(s_i, s_j), & i \neq j, \\ 0, & i = j, \end{cases} \quad \ell_i = \frac{1}{K(p, s_i)}. \end{aligned} \quad (5.31)$$

Then $n_+(\mathcal{B}) \leq 1$.

Proof. First fix a compactly supported measure $0 \leq \rho \leq q$. Choose a nonempty compact set $E \subset \mathbb{D}$ with $\rho(\mathbb{D} \setminus E) = 0$. For $s \in S^1$ and $x \in \mathbb{D}$, the Green formula gives, when $rs \neq x$,

$$G(rs, x) = \frac{1}{4\pi} \log \left(1 + \frac{(1 - r^2)(1 - |x|^2)}{|rs - x|^2} \right).$$

On any fixed compact interior set, $|rs - x|$ is bounded away from zero for r sufficiently close to one. Thus $\omega_{r,s}(x) = (1 - r^2)(1 - |x|^2)/|rs - x|^2$ tends uniformly to zero, and

$$\frac{G(rs, x)}{1 - r} = \frac{(1 + r)(1 - |x|^2)}{4\pi|rs - x|^2} \cdot \frac{\log(1 + \omega_{r,s}(x))}{\omega_{r,s}(x)}.$$

Since $\log(1+u)/u \rightarrow 1$ as $u \rightarrow 0$, this proves

$$\frac{G(rs, x)}{1-r} \longrightarrow P_s(x) = \frac{1-|x|^2}{2\pi|s-x|^2}, \quad (5.32)$$

uniformly on every compact subset of $\mathbb{D}$. For distinct $s, t \in S^1$, the same Green formula gives

$$G(rs, rt) = \frac{1}{4\pi} \log \left(1 + \frac{(1-r^2)^2}{r^2|s-t|^2} \right).$$

The fraction inside the logarithm tends to zero, and division by $(1-r)^2$, using $(1-r^2)^2/(1-r)^2 = (1+r)^2$, yields

$$\frac{G(rs, rt)}{(1-r)^2} \longrightarrow \frac{1}{\pi|s-t|^2} = N_0(s, t). \quad (5.33)$$

Each Poisson section P_s is bounded on E , so it belongs to $L^2(\rho)$. The uniform convergence in (5.32) gives

$$\left\| \frac{G(rs, \cdot)}{1-r} - P_s \right\|_\rho \leq \sqrt{\rho(E)} \sup_{x \in E} \left| \frac{G(rs, x)}{1-r} - P_s(x) \right| \longrightarrow 0. \quad (5.34)$$

Here $(I - T_\rho)^{-1}$ is a fixed bounded operator on $L^2(\rho)$, with norm at most $1/(1-d)$. Expanding its Neumann series and using equations (5.28) to (5.30) gives the identities

$$\begin{aligned} R_\rho(s, t) &= \langle P_s, (I - T_\rho)^{-1} P_t \rangle_\rho, \\ K_\rho(p, s) &= P_s(p) + \langle G(p, \cdot), (I - T_\rho)^{-1} P_s \rangle_\rho. \end{aligned} \quad (5.35)$$

All sections in these pairings belong to $L^2(\rho)$, including $G(p, \cdot)$ by the square integrability noted in section 5.3.

For $r > |p|$, the points $rs_1, \dots, rs_n, p$ are distinct. Writing out (5.9) at these points gives

$$\begin{aligned} \frac{M_\rho(rs_i, rs_j)}{(1-r)^2} &= \frac{G(rs_i, rs_j)}{(1-r)^2} \\ &\quad + \left\langle \frac{G(rs_i, \cdot)}{1-r}, (I - T_\rho)^{-1} \frac{G(rs_j, \cdot)}{1-r} \right\rangle_\rho \quad (i \neq j), \\ \frac{M_\rho(p, rs_i)}{1-r} &= \frac{G(p, rs_i)}{1-r} \\ &\quad + \left\langle G(p, \cdot), (I - T_\rho)^{-1} \frac{G(rs_i, \cdot)}{1-r} \right\rangle_\rho. \end{aligned} \quad (5.36)$$

By (5.34) and boundedness of the resolvent, the first pairing tends to $R_\rho(s_i, s_j)$ and the second tends to $K_\rho(p, s_i) - P_{s_i}(p)$, by (5.35). The direct terms tend to $N_0(s_i, s_j)$ and $P_{s_i}(p)$ by equations (5.32) and (5.33), using symmetry of G for the second term. Consequently

$$\begin{aligned} \frac{M_\rho(rs_i, rs_j)}{(1-r)^2} &\longrightarrow N_\rho(s_i, s_j) \quad (i \neq j), \\ \frac{M_\rho(p, rs_i)}{1-r} &\longrightarrow K_\rho(p, s_i). \end{aligned} \quad (5.37)$$

Let $Q_{\rho, r}$ be the reciprocal matrix of M_ρ sampled at $rs_1, \dots, rs_n, p$, with zero diagonal as in Definition 5.1. By Proposition 5.5, $n_+(Q_{\rho, r}) \leq 1$. Define

$$D_r = \text{diag}(\underbrace{1-r, \dots, 1-r}_{n \text{ entries}}, 1).$$

Since D_r is invertible, Sylvester's law of inertia gives $n_+(D_r Q_{\rho,r} D_r) \leq 1$. Its entries satisfy

$$\begin{aligned} (D_r Q_{\rho,r} D_r)_{ij} &= \frac{(1-r)^2}{M_\rho(rs_i, rs_j)} \quad (i \neq j, i, j \leq n), \\ (D_r Q_{\rho,r} D_r)_{i,n+1} &= \frac{1-r}{M_\rho(rs_i, p)} \quad (i \leq n). \end{aligned}$$

Every diagonal entry remains zero. The limits in (5.37) are strictly positive, since $N_\rho \geq N_0 > 0$ off the diagonal and $K_\rho(p, s) \geq P_s(p) > 0$. Thus those limits and symmetry imply the entrywise matrix convergence

$$\begin{aligned} D_r Q_{\rho,r} D_r &\longrightarrow \mathcal{B}_\rho = \begin{pmatrix} B_\rho & \ell_\rho \\ \ell_\rho^T & 0 \end{pmatrix}, \\ (B_\rho)_{ij} &= \begin{cases} 1/N_\rho(s_i, s_j), & i \neq j, \\ 0, & i = j, \end{cases} \quad (\ell_\rho)_i = \frac{1}{K_\rho(p, s_i)}. \end{aligned} \tag{5.38}$$

Since the matrices have fixed size, the bound $n_+ \leq 1$ passes to the limit. Hence $n_+(\mathcal{B}_\rho) \leq 1$ for every compactly supported $0 \leq \rho \leq q$.

Now choose $0 < r_1 < r_2 < \dots \uparrow 1$, set $F_m = \{|x| \leq r_m\}$, and define $d\rho_m = \mathbf{1}_{F_m} dq$. To apply monotone convergence, write the integrals in equations (5.28) and (5.29) against the fixed product measures formed from q . Their integrands are respectively the displayed products of G and P_s , multiplied by

$$\prod_{\ell=1}^k \mathbf{1}_{F_m}(x_\ell) \quad \text{and} \quad \prod_{\ell=0}^k \mathbf{1}_{F_m}(x_\ell).$$

All factors are nonnegative. Since $F_m \uparrow \mathbb{D}$, these indicator products increase pointwise to one. Monotone convergence therefore gives, for each fixed k ,

$$\mathcal{K}_{\rho_m, k}(p, s) \uparrow \mathcal{K}_{q, k}(p, s), \quad \mathcal{R}_{\rho_m, k}(s, t) \uparrow \mathcal{R}_{q, k}(s, t).$$

Applying monotone convergence to the nonnegative series in (5.30) yields

$$\begin{aligned} K_{\rho_m}(p, s) &\uparrow P_s(p) + \sum_{k=1}^{\infty} \mathcal{K}_{q, k}(p, s) = K(p, s), \\ R_{\rho_m}(s, t) &\uparrow \sum_{k=0}^{\infty} \mathcal{R}_{q, k}(s, t) = R(s, t), \\ N_{\rho_m}(s, t) &\uparrow N_0(s, t) + R(s, t) = N(s, t) \quad (s \neq t). \end{aligned} \tag{5.39}$$

These limits are exactly the kernels constructed for q in Section 4. Lemma 4.2 states that $K(p, s)$ is finite for $p \in \mathbb{D}$ and $s \in S^1$, and that $R(s, t)$ is finite for distinct $s, t \in S^1$. Thus the limiting K and N values used here are finite and strictly positive. Taking their reciprocals in (5.39) gives $\mathcal{B}_{\rho_m} \rightarrow \mathcal{B}$ entrywise, with the diagonal identically zero. Since $n_+(\mathcal{B}_{\rho_m}) \leq 1$, the same bound holds for $\mathcal{B}$, proving the lemma. $\square$

Proof of Theorem 5.1. For distinct $s_1, \dots, s_n \in S^1$, use the matrix $\mathcal{B}$ in (5.31). By lemma 5.6, $n_+(\mathcal{B}) \leq 1$. Each $K(p, s_i)$ is finite and strictly positive, so $\ell_i = 1/K(p, s_i) > 0$ and $\ell \neq 0$.

We first show that $v^T Bv \leq 0$ whenever $\ell^T v = 0$. Suppose instead that $\ell^T v = 0$ and $v^T Bv > 0$. Set $y_0 = \ell/(\ell^T \ell)$, so $\ell^T y_0 = 1$, and define

$$\hat{y} = y_0 - \frac{v^T B y_0}{v^T B v} v.$$

Then $v^T B \hat{y} = 0$ and $\ell^T \hat{y} = 1$. Choose $c \in \mathbb{R}$ so that $\hat{y}^T B \hat{y} + 2c > 0$. For the vectors $a_* = (v, 0)$ and $b_* = (\hat{y}, c)$,

$$\begin{aligned} a_*^T \mathcal{B} a_* &= v^T B v > 0, \\ b_*^T \mathcal{B} b_* &= \hat{y}^T B \hat{y} + 2c > 0, \\ a_*^T \mathcal{B} b_* &= v^T B \hat{y} + c \ell^T v = 0. \end{aligned}$$

The vectors are linearly independent, because $\ell^T v = 0$ while $\ell^T \hat{y} = 1$. The vanishing cross term shows that the quadratic form of $\mathcal{B}$ is positive definite on their two-dimensional span, contradicting $n_+(\mathcal{B}) \leq 1$. Therefore

$$v^T B v \leq 0 \quad \text{whenever } \ell^T v = 0. \quad (5.40)$$

Now let $b_1, \dots, b_n$ be real coefficients with $\sum_i b_i = 0$, and set $v_i = K(p, s_i) b_i$. Then

$$\ell^T v = \sum_i \frac{K(p, s_i) b_i}{K(p, s_i)} = 0,$$

and, by equations (5.1) and (5.31),

$$\sum_{i,j=1}^n b_i b_j D_p(s_i, s_j) = v^T B v \leq 0.$$

The diagonal terms vanish on both sides, and the inequality follows from (5.40). This proves conditional negative semidefiniteness for distinct points. For a finite family with repeated points, replace the coefficients at each distinct point by their sum. This preserves both the total coefficient sum and the quadratic expression, so the same conclusion holds. $\square$

6 Hilbert multipliers and the absence of critical points

Write $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and $S^1 = \partial \mathbb{D}$. We use the solution kernel K and the boundary kernel N constructed in section 4. section 5 proves that, for each $p \in \mathbb{D}$, the symmetric kernel

$$D_p(s, t) = \frac{K(p, s) K(p, t)}{N(s, t)} \quad (s \neq t), \quad D_p(s, s) = 0,$$

is conditionally negative semidefinite on S^1 . We now realize this kernel as squared distances in a real Hilbert space, following Schoenberg's classical correspondence [35]. The proof below gives the anchored Gram-kernel construction explicitly.

Lemma 6.1. *For every $p \in \mathbb{D}$, there exist a separable real Hilbert space $\mathcal{H}$ and an injective Lipschitz map $b : S^1 \rightarrow \mathcal{H}$ such that*

$$\|b(s) - b(t)\|_{\mathcal{H}}^2 = D_p(s, t) \quad (s, t \in S^1). \quad (6.1)$$

Proof. Fix $p \in \mathbb{D}$ and $s_0 \in S^1$, and define the anchored kernel

$$\Gamma(s, t) = \frac{1}{2}(D_p(s, s_0) + D_p(t, s_0) - D_p(s, t)).$$

To prove that Γ is positive semidefinite, take any $s_1, \dots, s_n \in S^1$ and $a_1, \dots, a_n \in \mathbb{R}$. Put $A = \sum_{i=1}^n a_i$ and $a_0 = -A$. Then $\sum_{i=0}^n a_i = 0$, and symmetry and $D_p(s_0, s_0) = 0$ give

$$\begin{aligned} \sum_{i,j=1}^n a_i a_j \Gamma(s_i, s_j) &= A \sum_{i=1}^n a_i D_p(s_i, s_0) - \frac{1}{2} \sum_{i,j=1}^n a_i a_j D_p(s_i, s_j) \\ &= -\frac{1}{2} \sum_{i,j=0}^n a_i a_j D_p(s_i, s_j) \geq 0. \end{aligned}$$

The last inequality follows from conditional negative semidefiniteness.

Let F be the real vector space of finitely supported functions on S^1 , and let $e_s \in F$ be the function equal to 1 at s and 0 elsewhere. Define a symmetric bilinear form on F by

$$B(f, g) = \sum_{s,t \in S^1} f(s)g(t)\Gamma(s, t).$$

These sums are finite, and positive semidefiniteness of Γ gives $B(f, f) \geq 0$. Set

$$Z = \{f \in F : B(f, f) = 0\}.$$

The Cauchy–Schwarz inequality for positive semidefinite bilinear forms gives

$$|B(f, g)|^2 \leq B(f, f)B(g, g).$$

Consequently, every element of Z is orthogonal to all of F . Thus Z is a linear subspace, and

$$\langle f + Z, g + Z \rangle = B(f, g)$$

is a well-defined inner product on F/Z .

Let $\mathcal{H}$ be the completion of this real inner product space, identify F/Z with its canonical image in $\mathcal{H}$, and define

$$b : S^1 \longrightarrow \mathcal{H}, \quad b(s) = e_s + Z.$$

Then $\langle b(s), b(t) \rangle_{\mathcal{H}} = \Gamma(s, t)$, so

$$\|b(s) - b(t)\|_{\mathcal{H}}^2 = \Gamma(s, s) + \Gamma(t, t) - 2\Gamma(s, t) = D_p(s, t).$$

Also $\Gamma(s_0, s_0) = 0$, so $b(s_0) = 0$.

For distinct $s, t \in S^1$, the kernel estimates from section 4 give

$$K(p, s) > 0, \quad K(p, t) > 0, \quad 0 < N(s, t) < \infty.$$

Hence

$$\|b(s) - b(t)\|_{\mathcal{H}}^2 = \frac{K(p, s)K(p, t)}{N(s, t)} > 0.$$

Thus $b(s) \neq b(t)$, proving injectivity.

Recall from section 4 that

$$N(s, t) = N_0(s, t) + R(s, t), \quad N_0(s, t) = \frac{1}{\pi|s - t|^2},$$

where $R(s, t) \geq 0$. Here N_0 is the boundary kernel for the disk with zero potential. Therefore, for $s \neq t$,

$$\begin{aligned} \|b(s) - b(t)\|_{\mathcal{H}}^2 &= \frac{K(p, s)K(p, t)}{N(s, t)} \\ &\leq \frac{K(p, s)K(p, t)}{N_0(s, t)} \\ &= \pi K(p, s)K(p, t)|s - t|^2 \\ &\leq \pi \|K(p, \cdot)\|_{\infty}^2 |s - t|^2. \end{aligned}$$

The estimate also holds when $s = t$, since both sides vanish. Taking square roots shows that b is Lipschitz, with

$$\text{Lip}(b) \leq \sqrt{\pi} \|K(p, \cdot)\|_{\infty}.$$

Finally, set

$$\mathcal{H}_0 = \overline{\text{span}_{\mathbb{R}} b(S^1)} \subseteq \mathcal{H}.$$

As a closed linear subspace of $\mathcal{H}$, the space $\mathcal{H}_0$ is itself a real Hilbert space with the inherited inner product. Viewing b as a map into $\mathcal{H}_0$ preserves its distances, injectivity, and Lipschitz bound. Since S^1 is separable and b is Lipschitz, $b(S^1)$ and its closed linear span $\mathcal{H}_0$ are separable. We therefore use $\mathcal{H}_0$ as the Hilbert space in the statement. $\square$

Proof of Theorem 1.1. Let V be the real Neumann eigenspace for the first positive eigenvalue λ_N , and take an arbitrary nonzero $u \in V$. Recall from Section 4 that

$$g(s) = \nabla u(\Phi(s)), \quad s \in S^1, \quad W_g = (\nabla u) \circ \Phi.$$

Here the gradient is taken in the Cartesian coordinates of Ω ; g is its boundary restriction parametrized by Φ , rather than the gradient of $u \circ \Phi$. Elliptic regularity [8, Chapter III, Theorems 6.4 and 6.5] gives $u \in C^\infty(\overline{\Omega})$, so g is smooth, and the Neumann condition makes $g(s)$ tangent to $\partial\Omega$ at $\Phi(s)$.

Suppose $\Phi(p)$ is a critical point of u , for some $p \in \mathbb{D}$. Choose b from lemma 6.1, and let b_j be its coordinates in a finite or countable orthonormal basis of $\mathcal{H}$. Each b_j is real Lipschitz, so the multiplier identity from section 4 applies to $b_j g$. For partial sums over the basis coordinates,

$$\begin{aligned} N(s, t) \sum_{j \leq m} |b_j(s) - b_j(t)|^2 |g(s)| |g(t)| \\ \leq N(s, t) D_p(s, t) |g(s)| |g(t)| \\ = K(p, s) K(p, t) |g(s)| |g(t)| \quad (s \neq t). \end{aligned}$$

The right-hand side is bounded and integrable on $S^1 \times S^1$. Parseval's identity and dominated convergence therefore give

$$\begin{aligned} \sum_j E(b_j g) &= \frac{1}{2} \iint_{S^1 \times S^1} K(p, s) K(p, t) g(s) \cdot g(t) ds dt \\ &= \frac{1}{2} \left| \int_{S^1} K(p, s) g(s) ds \right|^2 = \frac{1}{2} |W_g(p)|^2 = 0. \end{aligned} \tag{6.2}$$

Here ds, dt denote arclength on S^1 , and the diagonal has product measure zero. In the finite-dimensional case the sums simply terminate.

Every summand is nonnegative by Proposition 3.1. Thus $E(b_j g) = 0$ for every j . Its equality case, as expressed in Section 4, gives $v_j \in V$ such that

$$\nabla v_j(\Phi(s)) = b_j(s)g(s) \quad (s \in S^1).$$

Initially this is an equality of boundary traces, holding almost everywhere. Elliptic regularity gives $v_j \in C^\infty(\bar{\Omega})$, while $b_j g$ is continuous, so the equality holds at every boundary point. By the definition of g , we also have $\nabla u(\Phi(s)) = g(s)$ for every $s \in S^1$.

Set $S_g = \{s \in S^1 : g(s) \neq 0\}$. This set is open and nonempty. Indeed, if it were empty, both Cartesian derivatives of u would vanish on $\partial\Omega$, since $\Phi(S^1) = \partial\Omega$. For $k = 1, 2$, put $w_k = \partial_k u$. Smoothness up to the boundary and zero boundary values give $w_k \in H_0^1(\Omega)$. Differentiating the eigenfunction equation yields

$$-\Delta w_k = \partial_k(-\Delta u) = \lambda_N \partial_k u = \lambda_N w_k.$$

Testing with w_k and using the Dirichlet variational inequality gives

$$\lambda_D \int_{\Omega} w_k^2 \leq \int_{\Omega} |\nabla w_k|^2 = \lambda_N \int_{\Omega} w_k^2.$$

Since $\lambda_N < \lambda_D$, both w_k vanish. Connectedness of Ω then makes u constant, and $-\Delta u = \lambda_N u$ with $\lambda_N > 0$ forces $u = 0$, a contradiction.

Case 1: S_g is dense in S^1 . The set $b(S_g)$ cannot be contained in an affine line. Otherwise, density and continuity would place $b(S^1)$ in that closed line. But a continuous injective map from the circle into a line is impossible: its compact connected image would be a nondegenerate closed interval, and the map would be a homeomorphism onto that interval. Removing the preimage of an interior point would leave the circle connected but disconnect the interval.

Choose $s_1, s_2, s_3 \in S_g$ whose images under b are affinely independent. Write $x = b(s_2) - b(s_1)$ and $y = b(s_3) - b(s_1)$. These vectors are linearly independent. Choose a coordinate j with $x_j \neq 0$. Some coordinate k satisfies $x_j y_k - x_k y_j \neq 0$; otherwise $y_k = (y_j/x_j)x_k$ for every k , which would give $y = (y_j/x_j)x$. This nonzero determinant shows that $1, b_j, b_k$ are linearly independent on the three chosen points. Since $g(s_i) \neq 0$, the functions $g, b_j g, b_k g : S^1 \rightarrow \mathbb{R}^2$ are linearly independent as well.

Consequently, u, v_j, v_k are linearly independent: any relation $\alpha u + \beta v_j + \gamma v_k = 0$ would, after taking gradients and evaluating at $\Phi(s)$, give

$$\alpha g + \beta b_j g + \gamma b_k g = 0.$$

Independence of these boundary functions forces $\alpha = \beta = \gamma = 0$. This contradicts the bound $\dim_{\mathbb{R}} V \leq 2$ proved in section 2.

Case 2: S_g is not dense in S^1 . There is a nonempty open arc $I \subset S^1$ on which $g = 0$. Since S_g is nonempty and open, it contains two distinct points. Injectivity of b implies that some coordinate b_j takes different values at these points, and hence is nonconstant on S_g . Thus g and $b_j g$ are linearly independent: a relation $\alpha g + \beta b_j g = 0$ implies $\alpha + \beta b_j(s) = 0$ on S_g , and nonconstancy forces $\alpha = \beta = 0$. The corresponding eigenfunctions u, v_j are also linearly independent, since taking gradients and evaluating on the boundary would turn any relation between them into the same relation between g and $b_j g$.

Since $g = 0$ on I , the identities $\nabla u \circ \Phi = g$ and $\nabla v_j \circ \Phi = b_j g$ show that both gradients vanish on $\Phi(I)$. In particular, both tangential derivatives vanish along this connected boundary arc, so u and v_j are each constant there. Denote their respective constants by c_u, c_j . Choose $(\alpha, \beta) \neq (0, 0)$ such that $\alpha c_u + \beta c_j = 0$, and set $w = \alpha u + \beta v_j$. Then $w = 0$ on $\Phi(I)$. As $w \in V$, it satisfies the homogeneous Neumann condition $\partial_\nu w = 0$ on all of $\partial\Omega$, and hence on this arc.

The following construction is a local version of Filonov's zero-extension argument [20, Lemma, p. 414]. Take a small connected neighborhood U of an interior point of $\Phi(I)$ that meets no other boundary portion. Define $\tilde{w} = w$ on $U \cap \Omega$ and $\tilde{w} = 0$ on $U \setminus \Omega$. For any $\varphi \in C_c^\infty(U)$, Green's identity gives

$$\begin{aligned} \int_U \tilde{w}(\Delta\varphi + \lambda_N\varphi) dx &= \int_{U \cap \Omega} w(\Delta\varphi + \lambda_N\varphi) dx \\ &= \int_{U \cap \Omega} (\Delta w + \lambda_N w)\varphi dx \\ &\quad + \int_{U \cap \partial\Omega} (w \partial_\nu \varphi - \varphi \partial_\nu w) d\sigma = 0. \end{aligned}$$

The interior term vanishes because $w \in V$, and both boundary terms vanish because $w = \partial_\nu w = 0$ on $U \cap \partial\Omega$. There are no contributions from the edge of U , since φ has compact support in U . Thus $(\Delta + \lambda_N)\tilde{w} = 0$ distributionally in U .

Interior elliptic regularity makes $\tilde{w}$ smooth in U . Fact 2.7 then implies that it is real analytic. Since it vanishes on the exterior open set $U \setminus \bar{\Omega}$, the same fact gives $\tilde{w} = 0$ throughout U . In particular, w vanishes on the nonempty open subset $U \cap \Omega$. Since w satisfies $-\Delta w = \lambda_N w$ in the connected domain Ω , Fact 2.7 implies that $w = 0$ throughout Ω . But we chose $w = \alpha u + \beta v_j$ with $(\alpha, \beta) \neq (0, 0)$. Thus $\alpha u + \beta v_j = 0$ is a nontrivial linear relation between u and v_j , contradicting their linear independence.

Both cases contradict the assumed critical point. Hence ∇u is nowhere zero in Ω . Since u is continuous on the compact set $\bar{\Omega}$, the extreme value theorem guarantees that it attains its maximum and minimum there. Fermat's stationary-point theorem from elementary calculus says that a differentiable function has zero gradient at an interior local maximum or minimum. Indeed, restricting the function to each coordinate line through that point gives a one-variable local extremum, so each partial derivative is zero. Thus any interior point attaining either extremum of u would satisfy $\nabla u = 0$. We have proved that no such point exists, so both extrema are attained on $\partial\Omega$. Moreover, equality with either boundary extremum at an interior point would make that point a global, hence local, extremum. Therefore

$$\min_{\partial\Omega} u < u(x) < \max_{\partial\Omega} u \quad (x \in \Omega). \quad \square$$

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